MOBILE COMPUTING

Complete Examination Study Guide

College of Business Education | ITU07401

Questions & Model Answers | Past Papers Solved | Test Papers Solved

11 Course Modules — Arranged to Match Your Textbook
MODULE 1: The OSI Model MODULE 2: Introduction to Mobile Computing & MAC Protocols MODULE 3: GSM (2G) MODULE 4: GPRS (2.5G) MODULE 5: UMTS (3G) & 4G LTE MODULE 6: 5G Networks MODULE 7: Cloud Computing MODULE 8: Internet of Things (IoT) MODULE 9: Mobile IP MODULE 10: TCP/IP in Mobile Networks MODULE 11: Mobile Ad Hoc Networks (MANET)
EXAM SECTION: CBE Past Paper (July 2018) Fully Solved | Test Paper 2 (Feb 2026) Fully Solved | Exam Strategy & Quick-Reference Tables

📌 *How to Use: Every Q&A follows: Define → Explain → Real Example → Conclude. Green boxes = actual CBE exam questions. Answer keys give full mark-by-mark breakdowns.*

MODULE 1

The OSI Model

7-layer framework | Data encapsulation | TCP/IP comparison | Mobile network context

1.1 What is the OSI Model?

OSI (Open Systems Interconnection) is a 7-layer framework created by ISO in 1984 to standardize how data travels between computers on any network. Before OSI, different manufacturers' computers could not communicate with each other. Each layer has ONE specific job and communicates only with the layer directly above and below it.

Memory Trick (top→bottom): 'All People Seem To Need Data Processing'

Application → Presentation → Session → Transport → Network → Data Link → Physical

Layer #	Name	Job	Key Protocols	Simple Trick
7	Application	Network services directly to user apps.	HTTP, HTTPS, FTP, DNS, SMTP	The app you USE
6	Presentation	Translates, encrypts, and compresses data.	TLS/SSL, JPEG, MPEG	The TRANSLATOR
5	Session	Opens, manages, and closes sessions.	NetBIOS, RPC	The SESSION manager
4	Transport	Reliable end-to-end delivery, segmentation.	TCP, UDP	The DELIVERY guarantee
3	Network	Routes packets across networks using IP.	IP, ICMP, OSPF	The NAVIGATOR
2	Data Link	Node-to-node delivery using MAC addresses.	Ethernet, WiFi 802.11	The LOCAL postman
1	Physical	Transmits raw bits over physical medium.	Cables, radio waves	The WIRE / SIGNAL

1.2 OSI vs TCP/IP Model

OSI Layer	TCP/IP Layer	Protocols
Application (7)	Application	HTTP, HTTPS, FTP, DNS,

		SMTP, MQTT
Presentation (6)	Application	SSL/TLS, JPEG, MPEG
Session (5)	Application	NetBIOS, RPC, SIP
Transport (4)	Transport	TCP, UDP
Network (3)	Internet	IP, ICMP, OSPF, BGP
Data Link (2)	Network Interface	Ethernet, WiFi 802.11
Physical (1)	Network Interface	Copper, Fibre, Radio waves

Key distinction: OSI = 7-layer THEORETICAL model for learning. TCP/IP = 4-layer PRACTICAL model the internet actually uses.

1.3 OSI in Mobile Networks

Layer	Mobile Network Specifics
Physical (1)	Radio waves instead of cables. 2G=900/1800 MHz, 3G=2100 MHz, 4G=700–2600 MHz, 5G=up to 100 GHz.
Data Link (2)	Wireless MAC: TDMA (GSM), CDMA (3G), OFDMA (4G/5G). Manages shared radio channel and handover.
Network (3)	Mobile IP manages routing when device moves. IP address must follow the mobile device.
Transport (4)	TCP struggles with wireless — mistakes signal loss for congestion and reduces speed unnecessarily.
Session (5)	Handles session continuity across handovers — session must not drop when moving between towers.
Application (7)	Mobile-specific protocols: WAP, HTTP/2 (compressed for mobile), MQTT (lightweight for IoT).

Q1: Explain the role of the Transport Layer in the OSI model.

✅ **Answer — 4-step model answer**

Step 1 — DEFINE: The Transport Layer (Layer 4) is responsible for reliable end-to-end delivery of data between a source and destination device across a network.

Step 2 — EXPLAIN: It breaks large data into segments, assigns sequence numbers for correct reassembly, uses port numbers (HTTPS=443, HTTP=80) to direct data to the correct application, and uses TCP for guaranteed delivery or UDP for speed without guarantee.

Step 3 — REAL EXAMPLE: When a student downloads a university assignment PDF from Moodle, TCP breaks the file into hundreds of segments. If any segment is lost, TCP retransmits it. The student receives a complete, uncorrupted PDF — not a broken file.

Step 4 — CONCLUDE: The Transport Layer is essential because it ensures data sent over an unreliable network arrives correctly and completely, regardless of what happens in the network infrastructure below it.

Q2: Compare TCP and UDP.

✅ **Answer**

Feature	TCP	UDP
Connection	3-way handshake required before sending.	No connection — send immediately.
Reliability	Guaranteed — lost packets retransmitted.	Not guaranteed — lost = gone.
Speed	Slower (acknowledgments add overhead).	Faster — no waiting for ACKs.
Order	Correct order guaranteed.	May arrive out of order.
Use cases	File download, web browsing, banking, email.	Live video/audio streaming, voice calls, online gaming, DNS.
Real example	Downloading a PDF from Moodle.	Watching a live football match on YouTube.

📝 **Exam Tip:** OSI exam tip: 'CSMA/CD used in Ethernet = Layer 2. IP routing = Layer 3. TCP/UDP = Layer 4. TLS encryption = Layer 6.' Know which protocol belongs to which layer.

MODULE 2

Introduction to Mobile Computing

Definition | Characteristics | MAC Protocols | Hidden/Exposed Terminal | FDMA vs TDMA vs CDMA

2.1 What is Mobile Computing?

Mobile Computing is a technology that allows transmission of data, voice, and video via a computer or any other wireless-enabled device without having to be connected to a fixed physical link.

Real Example: A doctor in a moving ambulance accessing a patient's medical records on a tablet — moving, wireless, live data. That is mobile computing.

2.2 Types of Mobility

Mobility Type	What It Means	Real Example
User Mobility	User moves physically but keeps same service/identity.	Student moves from Dar es Salaam to Mwanza, still accesses university portal with same credentials.
Network Mobility	Switches between network types but keeps same session.	Phone switches from home WiFi to 4G as you leave — call doesn't drop.
Device Mobility	User switches devices but continues same task.	Start writing email on phone, finish it on laptop — same draft.
Session Mobility	Unfinished session moves from one device to another.	Pause a movie on phone, resume exactly where you left off on smart TV.
Bearer Mobility	Switches channel type when one is unavailable.	WhatsApp switches from WiFi calling to cellular if WiFi drops.
Host Mobility	Device acts as server and maintains reachable IP while moving.	A server on a truck that moves between cities but remains reachable.

Q3 [Exam]: Define Mobile Computing. Explain FOUR characteristics and TWO constraints.

✅ **Answer — CBE Q1 — 2+9 marks**

Mobile Computing is a technology that allows transmission of data, voice, and video via a computer or wireless-enabled device without a fixed physical link.

FOUR CHARACTERISTICS:

- Mobility — Users move between locations while remaining connected without interruption. Example: Sales executive checking emails while commuting by train.
- Portability — Devices are small and lightweight enough to carry anywhere. Example: Field engineer carrying a tablet to a construction site.
- Wireless Communication — Uses WiFi, 3G/4G/5G, Bluetooth. No physical cables. Example: Browsing internet on phone using 4G coverage.
- Connectivity — Device automatically switches between available networks. Example: Smartphone seamlessly switches from home WiFi to 4G as you leave home — video stream continues uninterrupted.

TWO CONSTRAINTS:

- Limited Resources — Less processing power, memory, storage, and battery than desktops. Example: GPS app drains phone battery in 3–4 hours.
- Variable Connectivity — Signal strength changes as user moves. Example: Clear video call at home breaks up in an area with weak signal.

2.3 MAC Protocols — Medium Access Control

MAC (Medium Access Control) defines rules by which multiple nodes share a single wireless channel. It is needed because the wireless channel is shared — simultaneous transmissions collide and are lost. Wireless nodes CANNOT detect collisions while transmitting (half-duplex operation).

2.3.1 Three Major Wireless MAC Challenges

1. Half-Duplex Operation: A node cannot transmit and receive simultaneously. Its own strong outgoing signal drowns incoming signals. This makes CSMA/CD (collision detection used in Ethernet) impossible. All wireless MAC focuses on collision AVOIDANCE.

2. Time-Varying Channel: Reflection, diffraction, and scattering cause multipath propagation. Received signal varies with time (fading). A node in fade may not detect an active transmission and wrongly assume the channel is free.

3. Burst Channel Errors: Wireless BER can reach 10^{-3} vs wired 10^{-6}. Errors occur in bursts during fades. Solutions: smaller packets, FEC (Forward Error Correction), retransmissions.

2.3.2 Hidden Terminal & Exposed Terminal Problems

Q4 [Exam]: With illustrations, explain how the Exposed Terminal Problem differs from the Hidden Terminal Problem.

✅ **Answer — CBE Q2a — 8 marks**

HIDDEN TERMINAL PROBLEM:

Setup: A ———► B ◄——— C (A and C cannot hear each other; both can reach B)

- A transmits to B. C cannot detect A's signal — A is out of C's range (hidden).

- C falsely senses channel as FREE and also transmits to B.
- COLLISION at B. Both transmissions corrupted. Neither A nor C knows.
- Problem: Collision and data loss. Solution: RTS/CTS — C hears CTS from B and stays silent.

EXPOSED TERMINAL PROBLEM:

Setup: A ◄——— B ———► C ———► D (D is out of B's range; C can hear B)

- B transmits to A. C wants to transmit to D (D outside B's range — no interference).
- C detects B's signal → wrongly assumes channel BUSY → C waits unnecessarily.
- C's transmission to D would NOT interfere. Problem: Wasted bandwidth.

KEY DIFFERENCE: Hidden = under-deferral (collision occurs). Exposed = over-deferral (no collision, but wasted opportunity).

2.3.3 RTS / CTS Mechanism

Q5 [Exam]: How are RTS and CTS used to avoid collision in multiple access?

✅ **Answer — CBE Q2c — 6 marks**

- Step 1 — RTS: Sender transmits short RTS to receiver [sender address, receiver address, duration].
- Step 2 — CTS: Receiver replies with CTS broadcast to ALL nearby devices [contains duration].
- Step 3 — NAV: ALL devices hearing CTS set NAV timer — stay silent for the reserved duration, including hidden terminals.
- Step 4 — Data Transmission: Sender transmits data confidently — channel is reserved.
- Step 5 — ACK: Receiver sends ACK. Channel released.

Why it prevents collisions: Hidden terminal C hears B's CTS → sets NAV → stays quiet even though it never heard A's RTS. Tiny 20-byte control frames collide instead of large data frames.

2.4 MAC Performance Metrics

Q6 [Exam]: Explain in detail how Delay and Throughput are used to compare MAC protocols.

✅ **Answer — CBE Q2b — 6 marks**

1. DELAY (Access Delay):

Definition: Average time from when a packet is queued until its transmission is fully complete. Includes: queuing delay, transmission delay, propagation delay, and retransmission delay. A protocol causing fewer collisions gives lower delay. Always compare protocols under IDENTICAL traffic conditions.

2. THROUGHPUT:

Definition: Fraction of channel capacity successfully used for useful data. Formula: $\eta = P / (T \times C)$ where P = packet size, T = elapsed time, C = channel capacity.

Trade-off: Reducing delay aggressively (transmit immediately) increases collisions → reduces throughput. A well-designed MAC must BALANCE both.

2.5 Multiple Access Technologies

Q7 [Exam]: Differentiate between FDMA and TDMA.

✅ **Answer — CBE Q1cii — 4 marks**

Feature	FDMA	TDMA	CDMA
How it works	Divides BANDWIDTH into frequency channels. Each user gets own dedicated frequency.	Divides TIME into slots. All users share same frequency, take turns per slot.	All users transmit simultaneously using unique codes to separate signals.
Simultaneous use	All transmit at same time on different frequencies.	Only one user per time slot.	All transmit at same time — code separation.
Synchronization	Not required between users.	Strict timing sync required.	Code synchronization required.
Power efficiency	Lower — idle channel wasted.	Higher — unused slots reallocated.	Highest — efficient spectrum use.
Analogy	Radio stations on different frequencies.	Meeting: 30 seconds each on same mic.	Room with conversations in different languages.
Used in	Early cellular; GSM gross division.	GSM (2G), PDC.	3G (UMTS/WCDMA).

GSM combines FDMA + TDMA: FDMA divides spectrum into 200 kHz channels, TDMA divides each into 8 time slots → 8 users per frequency.

2.6 WSN MAC Protocols — Energy Efficiency

Energy Waste Sources in Wireless Sensor Networks

Waste Source	Description
Idle Listening	Radio ON when no data expected — nearly same energy as active receive. Major WSN energy waster.
Collisions	Two signals at once → both lost. Transmission energy wasted for all colliding nodes.

Overhearing	Node receives packets NOT destined for it — wastes processing and energy.
Protocol Overhead	RTS/CTS control frames carry no data but cost energy — can be up to 70% of traffic.
Overmitting	Data sent to a node not ready to receive — wasted energy on useless transmissions.
Traffic Fluctuation	Fixed duty cycles mismatch variable traffic — energy wasted during low-traffic periods.

Q8: What is S-MAC? Explain its key components and limitations.

✅ **Answer**

S-MAC (Sensor-MAC) is an energy-efficient MAC for WSNs, inspired by IEEE 802.11, using periodic listen/sleep to reduce idle listening.

Four Key Components:

- Periodic Listen & Sleep: Alternates between fixed 115ms active periods and variable sleep — reduces idle listening.
- Time Synchronization: SYNC packets at start of each active period so neighbours share sleep schedules.
- Collision Avoidance: RTS/CTS mechanism handles hidden terminals.
- Overhearing Avoidance: Node hears RTS/CTS not addressed to it → sleeps for expected transmission duration.

Limitations: Poor scalability, fairness issues (data-heavy nodes can monopolize channel), does NOT adapt to traffic load.

Protocol	Type	Key Feature	Main Weakness
S-MAC	Contention (CSMA)	Periodic sleep/listen, RTS/CTS	Fixed active period — poor under variable load
T-MAC	Contention (CSMA)	Timeout-based early sleep — 5× less power under variable load	Early sleeping in multi-hop networks
B-MAC	Contention (LPL)	Low Power Listening with preamble sampling — flexible	Long preamble increases latency
LMAC	TDMA	Distributed slot selection — minimizes transceiver switches	Strict synchronization required
A-MAC	TDMA	Allows transmission to multiple destinations	Strict synchronization required
Z-MAC	Hybrid	Combines CSMA + TDMA advantages	Complex transition mechanisms

📝 **Exam Tip:** User/Host Mobility difference: User Mobility = PERSON continues their service (login follows them). Host Mobility = DEVICE maintains its network address using Mobile IP.

MODULE 3

GSM — Global System for Mobile Communication

2G Digital Standard | TDMA+FDMA | Architecture | Security | Services

3.1 What is GSM?

GSM (Global System for Mobile Communication) is a 2nd generation (2G) digital mobile telephony standard by ETSI. It replaced analog 1G systems with digital voice, SMS, and basic data using TDMA+FDMA. Operates at 900/1800 MHz (Europe/Africa/Asia) and 850/1900 MHz (North America). Introduced the SIM card concept.

Q9 [Exam]: What is GSM? In which frequency bands does it operate?

✅ **Answer — Q3a — 2 marks**

Band	Uplink	Downlink	Region
GSM-900	890–915 MHz	935–960 MHz	Europe, Africa, Asia, Australia
GSM-1800 (DCS)	1710–1785 MHz	1805–1880 MHz	High-density cities alongside GSM-900
GSM-850	824–849 MHz	869–894 MHz	North America, Latin America
GSM-1900 (PCS)	1850–1910 MHz	1930–1990 MHz	North America, Caribbean

East Africa primarily uses GSM-900 and GSM-1800 bands.

3.2 GSM Services

Q10 [Exam]: Differentiate between Tele Services and Supplementary Services in GSM.

✅ **Answer — Q3b — 4 marks**

Aspect	Tele Services	Supplementary Services
Definition	Primary end-to-end services delivering voice/text/data directly between users.	Optional features ON TOP of Tele Services — modify or enhance the basic experience.
Purpose	Provide the core communication service itself.	Modify, enhance, or control how the core service behaves.
Examples	Voice calls, Emergency calling	Call Forwarding, Call Hold, Call

	(999/112 — free, no SIM needed), SMS (160 chars), MMS, Fax.	Waiting, Multi-party (up to 6), Caller ID, Call Barring.
Standalone?	YES — works independently.	NO — cannot exist without a Tele Service.
In use	Calling a friend = Tele Service.	Putting first call on Hold to answer second = Supplementary Service.

3.3 GSM Architecture

Q11 [Exam]: With the help of GSM architecture, explain the basic parts of the Network Subsystem (NSS).

✅ **Answer — Q3c — 10 marks**

GSM architecture: Mobile Station (MS) ↔ Base Station Subsystem (BSS) ↔ Network Subsystem (NSS) ↔ External Networks

A) Mobile Station (MS):

- Mobile Equipment (ME): Physical handset identified by IMEI (15-digit serial number).
- SIM Card: Smart card with IMSI (subscriber identity), Ki (secret key), contacts, PIN protection.

B) Base Station Subsystem (BSS) — Radio layer:

- BTS (Base Transceiver Station): Radio transceivers and antennas for each cell.
- BSC (Base Station Controller): Controls multiple BTSs — handover, RF power, cell configuration.

C) Network Subsystem (NSS) — Core network:

Component	Full Name	Role	Analogy
HLR	Home Location Register	PRIMARY subscriber database — permanent data: MSISDN, IMSI, subscription, location, status.	National ID register.
MSC	Mobile Switching Center	Core telephone exchange — routes calls, billing (CDRs), signaling.	Telephone exchange operator.
VLR	Visitor Location Register	TEMPORARY database per MSC — subscriber data copied from HLR for fast local service. Deleted when	Hotel guest register.

		subscriber leaves.	
AUC	Authentication Center	Runs A3 (auth), A8 (key gen), A5 (encryption). Stores secret Ki — never transmitted over air.	Passport control officer.
EIR	Equipment Identity Register	IMEI database: White (allowed), Grey (suspect), Black (stolen/blocked).	Stolen vehicle database.

3.4 GSM Security

Q12: How does GSM security work? Explain A3, A5, and A8.

✅ **Answer**

Algorithm	Purpose	Input	Output
A3	Authentication	128-bit RAND challenge + 128-bit Ki (secret key)	32-bit SRES — network verifies identity by comparing SRES values
A8	Session Key Generation	128-bit RAND + 128-bit Ki	64-bit session key Kc — used by A5 for encryption
A5	Encryption (stream cipher)	Kc + frame number	Encrypted bitstream — A5/1=strong, A5/2=weak, A5/3=Kasumi (3G)

GSM also uses TMSI (Temporary Mobile Subscriber Identity) instead of IMSI over the air, protecting subscriber identity from eavesdroppers.

Mobile Telecommunications Services

3A.1 What Are Mobile Telecommunications Services?

Mobile Telecommunications Services are the communication capabilities offered to subscribers by a mobile network operator through the GSM infrastructure. The ITU and GSM standards formally classify these services into three categories — Bearer Services, Tele Services, and Supplementary Services — each serving a distinct role in the overall service architecture.

Core Formula

Mobile Telecom Service = Bearer Service (transport pipe) + Tele Service (user application) + Supplementary Service (optional feature that modifies or enhances the Tele Service).

Service Category	Purpose	Who Defines It	Examples
Bearer Services	Provide the raw data transport capability between two endpoints. Define HOW data moves, not what it is.	GSM/3GPP standards	Circuit-switched data (9.6 kbps), Packet-switched (GPRS), SMS transport layer
Tele Services	End-to-end communication services delivered directly to the user. Combine Bearer + Application layers.	Network operator + 3GPP	Voice calls, Emergency calls (112/999), SMS, MMS, Fax
Supplementary Services	Optional enhancements layered ON TOP of Tele Services. Cannot exist without a Tele Service.	Network operator (menu-driven)	Call Forwarding, Call Hold, Call Waiting, Multi-party, Caller ID, Call Barring

3A.1.1 Bearer Services — The Transport Layer

Bearer Services define the transport mechanism between the Mobile Station (MS) and the network — essentially the 'pipe' that carries data. They operate at Layers 1–3 of the OSI model and are transparent to the actual content being carried.

Bearer Type	Technology	Max Speed	Connection Type	Typical Use
Asynchronous Data	Circuit-switched	9.6 kbps	Dedicated connection	Dial-up internet, fax modems
Synchronous Data	Circuit-switched	9.6 kbps	Dedicated connection	Point-of-sale terminals

Packet Data (GPRS)	Packet-switched	171.2 kbps (theoretical)	Shared — billed per MB	Mobile email, web browsing
SMS Bearer	Signaling channel (SDCCH)	160 chars/message	Short-burst	Text messaging

3A.1.2 Tele Services — What Users Actually Use

Tele Services are complete end-to-end communication services that are directly visible to the subscriber. They use Bearer Services underneath but add application-layer functionality to create the full user experience. Each Tele Service has a standardised 2-digit service code.

Tele Service	Code	How It Works	Key Feature	Real Example (Tanzania)
Telephony (Voice)	11	Circuit-switched 64kbps full-duplex voice. Uses CODEC (GSM-FR, EFR, AMR) to compress voice into 13 kbps frames.	Real-time bidirectional voice	Calling family from Dar es Salaam to Mwanza — Vodacom GSM voice call
Emergency Calling	12	Free call to 112 or 999. Routed to nearest emergency centre. Works WITHOUT SIM card and even in another operator's coverage.	No SIM, no credit needed	Car accident on Morogoro Road — dial 112 from any GSM phone
SMS (Short Message)	21	160 alphanumeric characters via SMSC (Short Message Service Centre). Store-and-forward — held if recipient offline, delivered when back.	Asynchronous — sent even if recipient offline	Receiving M-Pesa transaction confirmation SMS from Vodacom
MMS (Multimedia)	61	Multimedia messages with images, audio, video. Uses WAP/HTTP over GPRS bearer. Handled by MMSC server.	Supports media attachments	Sending a birthday photo via MMS
Fax (Group 3)	31/32	Circuit-switched 9.6 kbps data call. Fax machine connects through terminal adapter.	Compatible with standard fax	Sending official documents from mobile fax adapter

3A.1.3 Supplementary Services — Enhancing the Experience

Supplementary Services are optional network features that modify or enhance a basic Tele Service. They are activated and deactivated by the subscriber through a Tele Service (usually voice calling) and CANNOT function independently.

Supplementary Service	What It Does	How to Activate (GSM Code)	Real Example

Call Forwarding Unconditional (CFU)	Diverts ALL incoming calls to another number immediately without ringing.	**21*+255712345678#	Forward calls to secretary when in a meeting
Call Forwarding Busy (CFB)	Diverts incoming calls ONLY when your line is already in a call.	**67*+255712345678#	Reroute to voicemail when already on a call
Call Forwarding No Reply (CFNR)	Diverts after 15–30 seconds of ringing with no answer.	**61*+255712345678**30#	Forward to colleague if you don't pick up in 30s
Call Waiting (CW)	Notifies you of a second incoming call while on a call. You can accept, reject, or ignore.	*43#	Hear a beep during a call — a second caller is waiting
Call Hold	Places current call on hold, allowing you to manage a second call or make an outgoing call.	Press Flash / GSM R key	Put wife on hold to accept urgent call from boss
Multi-party / Conference	Merges up to 5 participants in a single voice call (6 including originator).	Merge active + held calls	Project team conference call between Dar, Mwanza, and Arusha
Caller Line ID (CLIP)	Displays the caller's phone number on your screen before you answer.	Automatic (network must support)	See '+255754123456' on screen before answering
Call Barring — BAOC	Bars ALL outgoing calls from the phone.	#33*PIN#	Company phone — prevent personal calls
Calling Line ID Restriction (CLIR)	Hides YOUR number from the person you are calling. Shows 'Private' or 'Unknown'.	#31# before the number	Dial #31#+255754123456 — your number is hidden

■ [CBE Q3b — 4 marks] Differentiate between Tele Services and Supplementary Services in GSM.

■ Model Answer:

Step 1 — Define: Tele Services are the PRIMARY end-to-end communication services that deliver actual voice/text/data directly between users. Supplementary Services are OPTIONAL enhancements that modify or add features to Tele Services.

Step 2 — Explain key differences: Tele Services are standalone — they work independently (e.g., making a voice call). Supplementary Services CANNOT work alone — they require an active or subscribed Tele Service (e.g., Call Waiting requires a Voice Call Tele Service to exist). Supplementary Services change HOW the Tele Service behaves.

Step 3 — Real Example: Calling a friend = Tele Service (Telephony, code 11). While on that call, receiving notification that a second caller is waiting = Supplementary Service (Call Waiting). The Tele Service provides the call; the Supplementary Service enhances the experience of managing it.

Step 4 — Conclude: The distinction is architectural — Tele Services form the foundation of user communication in GSM, while Supplementary Services layer optional intelligence on top without changing the core transport mechanism.

■ ***Exam Tip:*** *Memory shortcut: 'Tele = the thing itself (THE CALL). Supplementary = the settings around it (what HAPPENS to the call).' In the exam, always note that Supplementary Services CANNOT exist without a Tele Service — this single sentence earns a guaranteed mark.*

3A.2 GSM Architecture — Complete Breakdown

GSM architecture is divided into four major functional areas interconnected by standardised interfaces. Understanding each component, its role, and the interfaces between them is essential for CBE examinations. The diagram below shows the complete architecture.

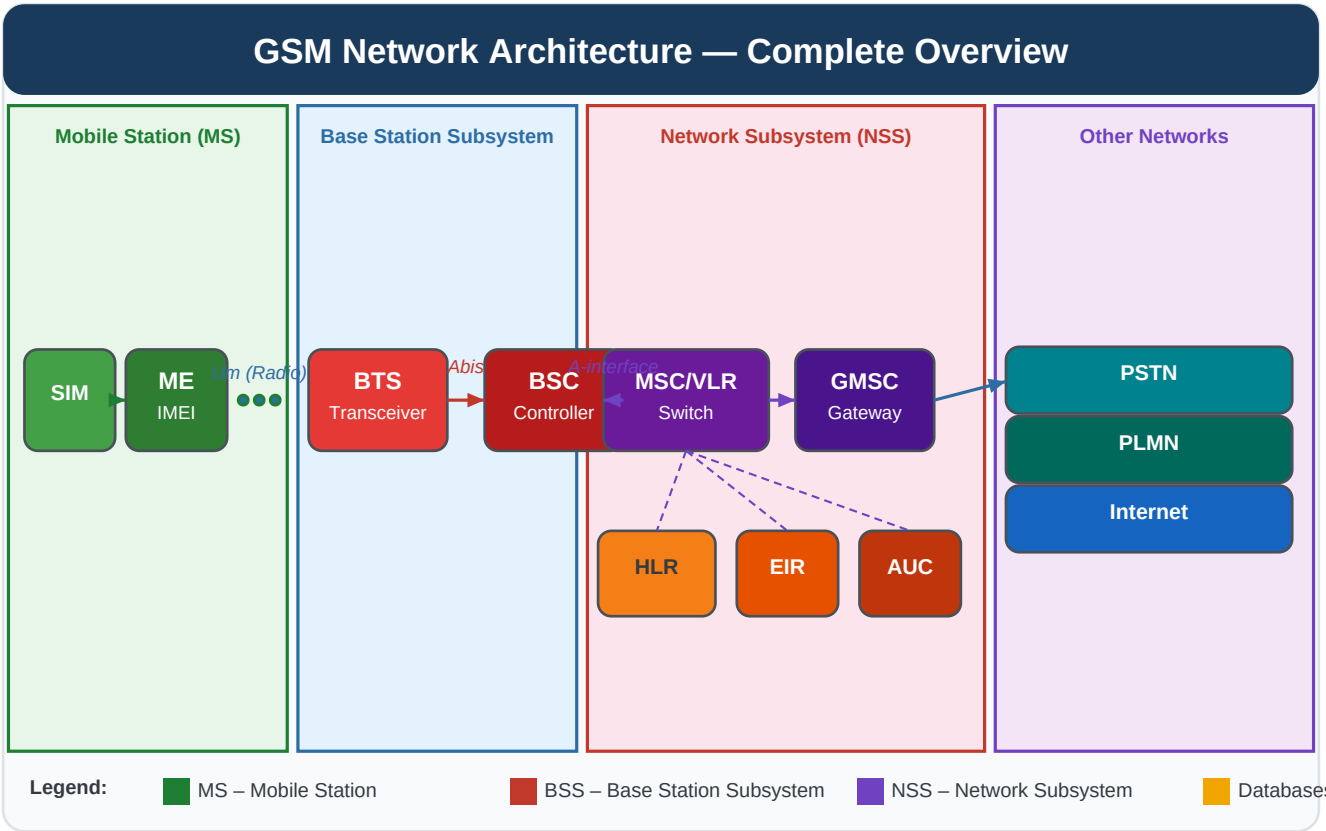

Figure 3A.1 — GSM Architecture: MS ↔ BSS ↔ NSS ↔ External Networks with all interfaces

3A.2.1 Zone 1 — Mobile Station (MS)

Component	Full Name	Identifier Used	Function	Key Detail
ME	Mobile Equipment	IMEI (15 digits)	The physical handset — radio transceiver, display, processor, battery.	IMEI = International Mobile Equipment Identity. Unique per device (not per subscriber). Can be blacklisted if stolen via EIR.
SIM	Subscriber Identity Module	IMSI (up to 15 digits)	Smart card containing subscriber identity, secret key Ki, and authentication algorithms A3/A8.	Ki is NEVER transmitted over the air. SIM stores TMSI (temp ID), last 5 dialled numbers, contacts, and PIN/PUK codes.

MS = ME + SIM

The Mobile Station is the COMBINATION of the physical device (ME) and the subscriber identity card (SIM). You can move a SIM between phones — your subscription follows the SIM, not the phone. The ME is identified by IMEI; the subscriber is identified by IMSI on the SIM.

3A.2.2 Zone 2 — Base Station Subsystem (BSS)

The BSS handles all radio-related functions — it is the bridge between the mobile handset and the wired core network. It contains two distinct elements connected by the Abis interface.

Component	Full Name	Connected To	Primary Functions	Analogy
BTS	Base Transceiver Station	Antenna tower → air interface to MS; Abis interface → BSC	• Transmits and receives radio signals to/from MS • One BTS covers one CELL (radius: 35 km max in rural areas, ~500m in urban) • Handles channel coding, interleaving, burst formatting, frequency hopping	Mobile phone tower — the antenna mast you see in the street
BSC	Base Station Controller	Multiple BTSs via Abis; MSC via A-interface	• Controls multiple BTSs (typically 10–100 per BSC) • Manages radio resource allocation (assigns channels to calls) • Handles INTRA-BSC handover (between BTSs it controls) • Controls transmission power of BTSs and MSs • Performs speech transcoding via TRAU	Traffic control room — manages which tower handles which call

3A.2.3 Zone 3 — Network Subsystem (NSS) — The Core Network

The NSS is the intelligence layer of GSM. It handles all switching, subscriber management, authentication, and interconnection with external networks. Every subscriber database, call routing decision, and security check passes through the NSS.

Component	Full Name	Primary Role	Key Stored Data	Analogy
MSC	Mobile Switching Centre	Core telephone exchange that ROUTES all calls. Handles call setup, teardown, billing (CDRs), and SMS relay. Every call passes through the MSC.	Current VLR reference, billing records (CDRs), connected BSC list	Head post office — routes every letter/call to correct destination
VLR	Visitor Location Register	TEMPORARY subscriber database co-located with MSC. When you enter an MSC's coverage area, your data is copied from HLR to VLR for fast local service. Deleted when you leave.	IMSI, MSISDN, TMSI, current LA (Location Area), subscribed services, authentication parameters	Hotel guest register — temporary record of visitors currently in the area

HLR	Home Location Register	PERMANENT master subscriber database for the home network. One per operator (or per region). Always holds your current VLR/MSC address so incoming calls can find you.	IMSI, MSISDN (phone number), subscription profile, current SGSN/MSC/VLR address, roaming restrictions	National ID database — permanent record of every subscriber
AUC	Authentication Centre	Generates authentication triplets (RAND, SRES, Kc) using Ki. Runs A3 (authentication), A8 (key generation). Physically secured — never transmits Ki.	Ki (secret key per subscriber), A3/A8 algorithm implementations, triplet generation log	Passport control — verifies identity without revealing secret documents
EIR	Equipment Identity Register	Database of IMEI numbers. Classifies phones as: WHITE (allowed), GREY (under investigation/monitoring), BLACK (stolen/defective — call rejected).	White/Grey/Black IMEI lists, device make/model data	Stolen vehicle database — checks if device is allowed on the network
GMSC	Gateway MSC	Entry/exit point for calls to/from external networks (PSTN, PLMN, Internet). Queries HLR to find the correct MSC for an incoming call, then routes it.	Routing tables for external networks, interconnect billing records	International airport — gateway between local and international networks

■ [CBE Q3c — 10 marks] With the help of GSM architecture, explain the basic parts of the Network Subsystem (NSS).

■ Full Model Answer — 6 NSS Components:

MSC (Mobile Switching Centre): The MSC is the HEART of the NSS — a digital telephone exchange that routes all voice calls and SMS. It connects the BSS (radio side) to the HLR, VLR, and external networks. Every call setup and teardown is controlled by the MSC. It also generates Call Detail Records (CDRs) for billing. Real example: When you call a Vodacom subscriber in Mwanza from Dar es Salaam, the Dar MSC sets up the circuit to Mwanza's MSC.

VLR (Visitor Location Register): The VLR is a TEMPORARY database that stores subscriber information for all MSs currently served by its MSC. When you switch on your phone in a new area, the MSC's VLR requests your profile from your HLR and stores a local copy. This eliminates the need to query the HLR for every call. When you leave the area, the VLR record is deleted. Real example: A Dar es Salaam subscriber travelling to Mwanza — Mwanza's VLR holds a copy of their data for fast local call handling.

HLR (Home Location Register): The HLR is the PERMANENT master database of all subscribers belonging to the operator. It stores the subscriber's MSISDN (phone number), IMSI, full subscription profile, roaming restrictions, and the address of the VLR currently serving the subscriber. The HLR is updated every time a subscriber moves to a new VLR area (Location Update). Real example: Vodacom Tanzania's HLR holds records for every Vodacom subscriber — it always knows which MSC/VLR is currently serving you, enabling incoming calls to route correctly.

AUC (Authentication Centre): The AUC is responsible for SECURITY. It stores the secret key Ki for each subscriber (same Ki as on the SIM) and generates authentication triplets (RAND, SRES, Kc) for the MSC to use. Ki NEVER leaves the AUC or SIM — authentication is proven by both sides computing the same SRES from the same RAND + Ki. Real example: When you switch on your phone, AUC generates a challenge (RAND) → your SIM computes SRES using Ki → network verifies → your identity is confirmed without

exposing Ki.

EIR (Equipment Identity Register): The EIR maintains a database of IMEI numbers (device identifiers) in three lists: WHITE LIST (authorised devices — calls permitted), GREY LIST (suspect devices — monitored but not blocked), BLACK LIST (stolen/faulty/barred devices — calls rejected). Real example: If your phone is stolen in Dar es Salaam and reported to the operator, your IMEI is moved to the BLACK LIST — the thief cannot make calls on ANY Tanzanian network even with a different SIM card.

GMSC (Gateway Mobile Switching Centre): The GMSC is the ENTRY POINT for calls arriving from external networks (PSTN, other PLMNs, internet). When someone calls your mobile number, PSTN routes to GMSC → GMSC queries HLR to find current MSC → GMSC routes call to that MSC. It also handles interconnect billing with external operators. Real example: When a landline calls your mobile number, the call enters GSM through the GMSC, which queries the HLR to find which MSC currently holds your VLR record, then routes the call accordingly.

Zone 4 — External Networks

Network	Full Name	Connection Point	What It Provides
PSTN	Public Switched Telephone Network	GMSC via ISUP/TUP	Traditional landline telephone system — calls from fixed phones to mobiles
PLMN	Public Land Mobile Network	GMSC (inter-operator interface)	Other mobile networks — enables roaming and inter-network calls
Internet	IP Network	GGSN (GPRS) / PGW (4G)	Data connectivity — web, email, app services

3A.3 GSM Interfaces — How the Components Talk to Each Other

Each connection between GSM components uses a defined, standardised interface. Knowing the interface names is important for exam questions on GSM architecture.

Interface	Between	Protocol	What It Carries	Key Characteristic
Um	MS ↔ BTS	GSM Radio (TDMA)	Voice, data, control signalling over the air	Wireless — radio interface. Uses TDMA frames. Susceptible to fading and interference.
Abis	BTS ↔ BSC	PCM/TDM (E1/T1)	64 kbps PCM voice channels + signalling	Wired or microwave. Carries multiple simultaneous calls between BTS and controller.
A	BSC ↔ MSC	SS7 (BSSAP)	Call setup signalling + user traffic	Standardised — allows multi-vendor BSS and MSC. Carries circuit-switched voice.
B	MSC ↔ VLR	MAP (internal)	Subscriber data queries and updates	Usually internal — MSC and VLR often co-located on same hardware.
C	MSC ↔ HLR (via GMSC)	MAP over SS7	Routing enquiries for incoming calls	GMSC queries HLR here to find which MSC serves the called subscriber.
D	HLR ↔ VLR	MAP over SS7	Location updates, subscriber profile transfer	Used during registration — VLR tells HLR 'subscriber is now here'; HLR sends profile.
E	MSC ↔ MSC	MAP/ISUP	Inter-MSC handover coordination + call routing	Used when subscriber moves between areas served by different MSCs.
F	MSC ↔ EIR	MAP over SS7	IMEI check requests	MSC queries EIR at call setup to verify device is not blacklisted.
G	VLR ↔ VLR	MAP over SS7	TMSI reallocation during inter-VLR movement	Transfers temporary identity information when subscriber moves between VLR areas.
Gs	VLR ↔ SGSN	MAP over SS7	GPRS/circuit-switched coordination	Coordinates paging between GSM and GPRS when subscriber is GPRS-attached.
Gr	SGSN ↔ HLR	MAP over SS7	GPRS subscriber registration and profile	SGSN requests subscriber data from HLR when MS performs GPRS attach.

■ ***Exam Tip:*** *For exam questions: 'Um = air (wireless). A-interface = BSC to MSC. D-interface = HLR to VLR. These three are the most commonly tested. Also remember: SS7 / MAP is the signalling protocol used on all the wired interfaces within the NSS.'*

3A.4 GSM Logical Channels

GSM uses a sophisticated channel structure to carry different types of information over the Um air interface. Logical channels are defined by their PURPOSE, not their physical frequency. They are divided into two main categories: Traffic Channels (TCH) and Control Channels (CCH).

Traffic Channels (TCH) — Carry Actual Voice / Data

Channel	Type	Data Rate	Used For
TCH/FS	Full-rate speech	22.8 kbps raw (13 kbps net)	Standard voice calls — one full TDMA slot
TCH/HS	Half-rate speech	11.4 kbps raw (6.5 kbps net)	Doubles capacity — two calls per slot at lower quality
TCH/F9.6	Full-rate data	9.6 kbps	Circuit-switched data calls (fax, dial-up)
TCH/F14.4	Enhanced full-rate data	14.4 kbps	Higher-speed circuit-switched data

Control Channels (CCH) — Carry Signalling and Management

Channel	Abbreviation	Direction	Purpose
Broadcast Control Channel	BCCH	BTS → MS (downlink only)	Broadcasts cell identity, network info, frequency list. MS reads this to camp on the cell.
Frequency Correction Channel	FCCH	BTS → MS	Carries a pure frequency tone so MS can synchronise its oscillator to the BTS.
Synchronization Channel	SCH	BTS → MS	Carries BSIC (Base Station Identity Code) and TDMA frame number for timing sync.
Random Access Channel	RACH	MS → BTS (uplink only)	MS sends an access burst to request a dedicated channel. First message MS ever sends.
Paging Channel	PCH	BTS → MS	Network pages MS when there is an incoming call or SMS. MS listens constantly.
Access Grant Channel	AGCH	BTS → MS	Network responds to RACH — assigns a dedicated channel (SDCCH) to the MS.
Standalone Dedicated Control Channel	SDCCH	Bidirectional	Used for call setup, SMS, authentication, and location update before TCH is assigned.
Slow Associated Control Channel	SACCH	Bidirectional	Sent alongside TCH — carries measurement reports (signal strength) and power control commands.
Fast Associated Control Channel	FACCH	Bidirectional	Urgent in-call signalling (e.g., handover commands). Steals frames from TCH — audible as click.

3A.5 GSM Call Flow — Mobile-Originated Call

Understanding the step-by-step sequence of a mobile-originated call demonstrates how all GSM components interact. Each step uses specific channels and interfaces.

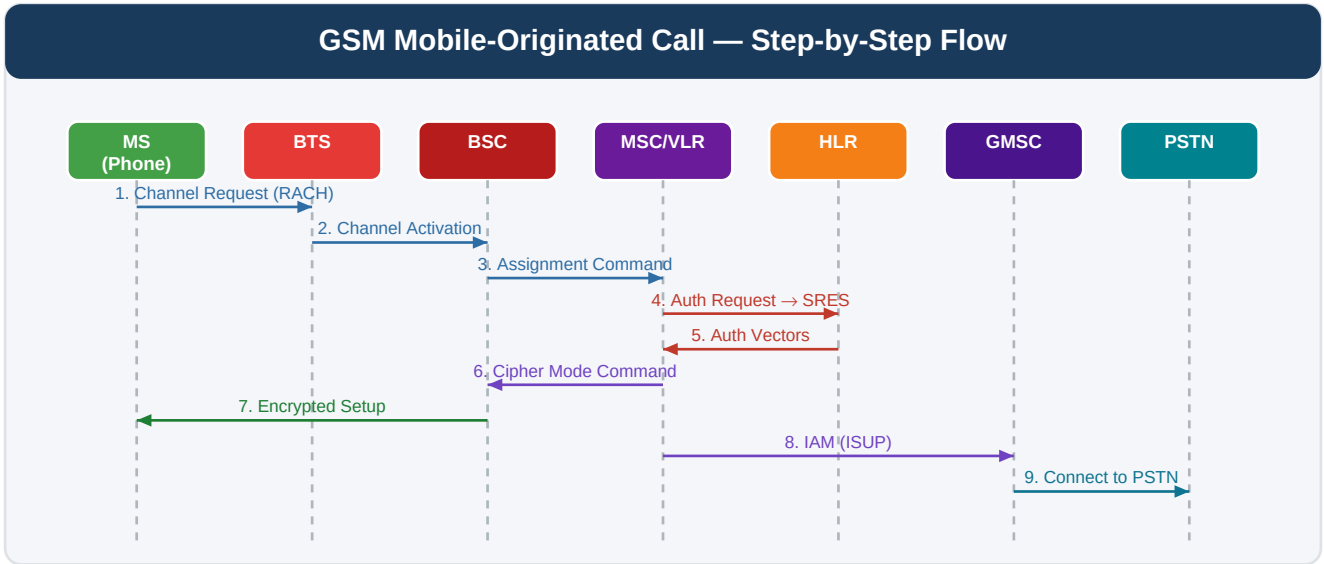

Figure 3A.2 — GSM Mobile-Originated Call Sequence (simplified)

Step	Message / Action	Channel / Interface	What Happens
1	Channel Request (RACH burst)	Um — RACH	MS sends random access burst to BTS requesting a dedicated channel.
2	Channel Activation	Abis	BSC instructs BTS to activate an SDCCH for the MS.
3	Immediate Assignment (AGCH)	Um — AGCH	BSC/BTS assigns SDCCH to MS — MS now has a dedicated signalling channel.
4	Authentication Request	Um — SDCCH → A → D	MSC requests triplet from VLR/AUC. Sends RAND to MS. MS SIM computes SRES via A3 algorithm.
5	Authentication Response	Um — SDCCH	MS sends SRES back. MSC compares with expected SRES — if match, MS is authenticated.
6	Cipher Mode Command	Um — SDCCH	MSC instructs BSC to start encryption using Kc (generated by A8). A5 algorithm encrypts all traffic.
7	CM Service Request + Call Setup	Um — SDCCH (now encrypted)	MS sends the dialled number to MSC. MSC assigns a TCH (traffic channel) for the voice call.
8	IAM (Initial Address Message)	A → E → ISUP	MSC sends IAM through GMSC toward PSTN or destination MSC with the called number.
9	Call Connect + TCH Assignment	Um — FACCH/TCH	When called party answers, MSC sends assignment — TCH carries the live voice conversation.

3A.6 GSM Security — Authentication, Encryption & Identity Protection

GSM introduced the first systematic security architecture in mobile telephony. It addresses three threats: subscriber impersonation, eavesdropping on calls, and tracking of subscriber identity. Three cryptographic algorithms — A3, A8, and A5 — form the security core.

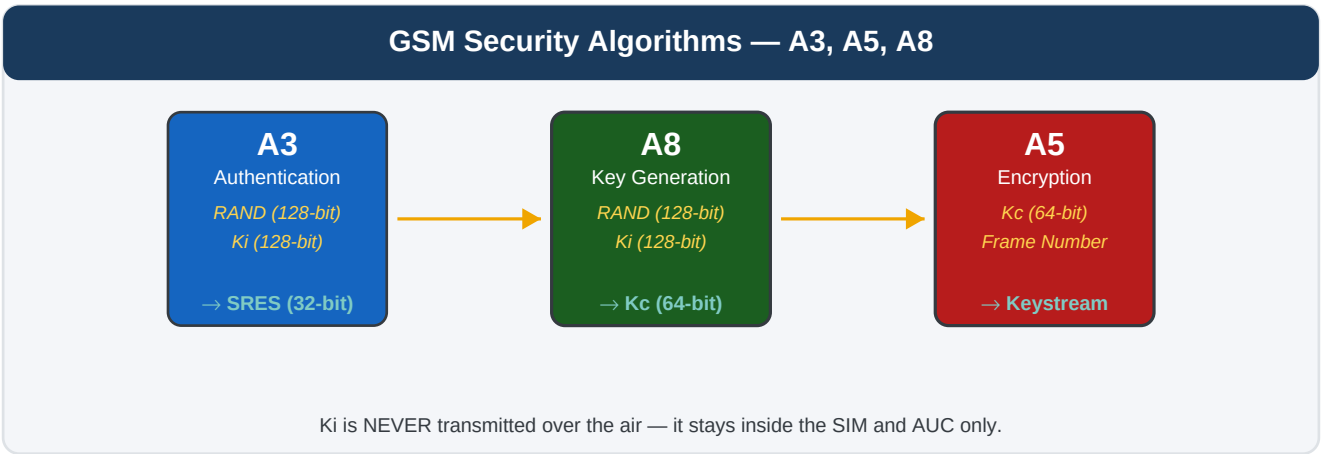

Figure 3A.3 — GSM Security: A3 authenticates, A8 generates the key, A5 encrypts.

Algorithm	Name	Location	Input	Output	Purpose
A3	Authentication Algorithm	SIM + AUC	Ki (128-bit secret key) + RAND (128-bit random challenge)	SRES (32-bit Signed Response)	Proves subscriber is genuine. Both SIM and AUC compute SRES independently — match = authentic.
A8	Key Generation Algorithm	SIM + AUC	Ki (128-bit) + RAND (128-bit)	Kc (64-bit session key)	Generates the encryption key Kc that will be used by A5 to encrypt the call.
A5	Stream Cipher (Encryption)	MS + BTS	Kc (64-bit) + TDMA frame number	Encrypted keystream	Encrypts voice/data over the Um interface. A5/1 = strong (West), A5/2 = weak (deprecated), A5/3 = Kasumi (3G-era).

3A.6.1 TMSI — Protecting Subscriber Identity Over the Air

To prevent an attacker from tracking a subscriber by their IMSI (which is permanent and linked to their identity), GSM uses TMSI (Temporary Mobile Subscriber Identity). The TMSI is a random 4-byte number assigned by the VLR and changes frequently. Over the air, the network pages by TMSI — not IMSI. IMSI is only used when TMSI is unknown (e.g., first registration or TMSI allocation failure).

Identifier	Full Name	Length	Changes?	Transmitted Over Air?	Purpose
IMSI	International Mobile Subscriber Identity	Up to 15 digits	NEVER (permanent per SIM)	Only at first registration or TMSI failure	Master subscriber identity — links to HLR record
MSISDN	Mobile Station ISDN Number	Up to 15 digits	Can change (number portability)	Never (internal to network)	The phone NUMBER that people dial
IMEI	International Mobile Equipment Identity	15 digits	Never (assigned at manufacture)	At call setup (for EIR check)	Device identity — identifies the handset, not the subscriber

TMSI	Temporary Mobile Subscriber Identity	32 bits (4 bytes)	Regularly (per VLR area)	Yes — used instead of IMSI	Privacy protection — temporary alias for IMSI over the air
LAI	Location Area Identity	MCC+MNC+LAC	When MS moves to new LA	Broadcast on BCCH	Identifies the location area — used for paging

■ ***Exam Tip:*** *Security exam shortcut: 'A3 = AUTHENTICATION (Are you who you say you are?). A8 = KEY (gives us the key to lock the conversation). A5 = ACTUAL ENCRYPTION (locks it). Ki never travels on the air — this is the single most important GSM security fact.'*

3A.7 GSM Handover — Maintaining Calls During Movement

Handover (also called Handoff) is the process of transferring an ongoing call from one BTS/cell to another as the subscriber moves. GSM handover is NETWORK-CONTROLLED — the network (BSC/MSC) makes the decision based on signal measurements, not the phone.

Handover Type	Controlled By	When It Occurs	Steps Involved
Intra-BSC Handover (Internal)	BSC	MS moves between two BTSs controlled by the SAME BSC	1. MS reports poor signal on SACCH 2. BSC selects better BTS 3. BSC activates channel on new BTS 4. MS switches — MSC not involved 5. Old channel released
Inter-BSC / Intra-MSC Handover	MSC	MS moves between BTSs under DIFFERENT BSCs, same MSC	1. Source BSC requests handover from MSC 2. MSC contacts target BSC 3. Target BSC activates channel 4. MS switches 5. Source BSC resources released
Inter-MSC Handover	Two MSCs + HLR	MS moves to a completely different MSC area	1. Anchor MSC (old) sends handover request to Relay MSC (new) 2. New MSC sets up new call leg 3. MS switches 4. HLR updated with new location

Handover Decision Criteria

The MS continuously measures signal strength and quality on the SACCH channel and reports to the BSC every 480 ms. The BSC also instructs the MS to measure neighbouring cells. Handover is triggered when: (1) received signal drops below threshold, (2) bit error rate (BER) exceeds threshold, (3) distance to BTS exceeds maximum timing advance (35 km), or (4) load balancing is needed.

3A.8 GSM Location Management

GSM must always know the approximate location of each subscriber so it can route incoming calls and page the correct base stations. This is achieved through a hierarchical location area system.

Concept	Definition	Typical Size	How Updated	Why Important
Cell	Area covered by one BTS antenna sector. Smallest unit.	500m (urban) to 35 km (rural)	Always known when call is active	Routing of active calls
Location Area (LA)	Group of cells served by one or more BTSs. Identified by LAI.	Dozens to hundreds of cells	MS sends Location Update to MSC when entering new LA	Limits paging area — network pages ALL BTSs in LA for incoming call

MSC/VLR Area	All LAs served by one MSC. Identified by MSC address.	Entire city or region	VLR updated when MS enters area	Routing of incoming calls to correct MSC
HLR / PLMN Area	Entire home network operator.	Entire country	HLR updated with current VLR address	Enables roaming and incoming calls from any network

Location Update Procedure:

- MS moves into a new Location Area — reads new LAI from BCCH.
- MS compares new LAI with stored LAI — if different, triggers Location Update.
- MS sends Location Updating Request to BSS → MSC → new VLR.
- New VLR registers MS and requests subscriber data from HLR (via D-interface).
- HLR updates the pointer to the new VLR address and sends profile.
- HLR sends Cancel Location to OLD VLR — old VLR deletes the record.
- New VLR allocates a new TMSI and sends Location Updating Accept to MS.
- MS stores new TMSI and new LAI — ready for service in new area.

Module 3A — Quick Reference: Key Numbers & Facts

Fact	Value / Detail
GSM generation	2G (Second Generation)
GSM standard body	ETSI (European Telecommunications Standards Institute)
Multiple access method	TDMA + FDMA (combined)
FDMA channel width	200 kHz per carrier frequency
TDMA slots per carrier	8 time slots (8 users per 200 kHz channel)
GSM frequency bands	GSM-900, GSM-1800, GSM-850, GSM-1900
East Africa primary bands	GSM-900 (890–960 MHz) and GSM-1800 (1710–1880 MHz)
Voice codec (full rate)	GSM-FR: 13 kbps net; EFR: 12.2 kbps; AMR: adaptive
SMS maximum length	160 alphanumeric characters
BTS maximum range	35 km (rural), ~500 m (urban dense)
Authentication algorithm	A3 (SRES output: 32 bits)
Key generation algorithm	A8 (Kc output: 64 bits)
Encryption algorithm	A5/1 (strong), A5/2 (weak/deprecated), A5/3 (Kasumi)
IMSI length	Up to 15 digits (MCC + MNC + MSIN)
IMEI length	15 digits

TMSI length	32 bits (4 bytes)
GSM circuit-switched voice data rate	64 kbps raw, 22.8 kbps after error correction, 13 kbps net
Supplementary Services max multi-party	6 participants (5 + originator)

■ ***Exam Tip:*** *Final exam strategy: For ANY architecture question, draw the four zones (MS / BSS / NSS / External Networks) and label all components. Then describe data flow left-to-right. For security, always mention Ki never leaves AUC/SIM. For services, always distinguish Tele (standalone) vs Supplementary (requires Tele Service).*

MODULE 4

GPRS — General Packet Radio Service

2.5G | Packet-switched data over GSM | SGSN & GGSN | Mobility Management | QoS

4.1 What is GPRS?

GPRS (General Packet Radio Service) is a packet-switched data service layered on top of GSM (2.5G). Unlike GSM's circuit-switched voice, GPRS is packet-switched — users are billed per data VOLUME rather than connection time. Maximum theoretical speed: 171.2 kbps; realistic: 5–40 kbps. Uses IP and X.25 protocols.

Aspect	GSM	GPRS
Switching	Circuit-switched (dedicated connection)	Packet-switched (shared resources)
Billing	Time-based (per minute)	Volume-based (per MB)
Speed	~9.6 kbps	Theoretical 171.2 kbps; realistic 5–40 kbps
Connection	Resources reserved for full session	Allocated only when data is being sent
Network Additions	None beyond GSM	Adds SGSN and GGSN to GSM core

4.2 GPRS Architecture — SGSN & GGSN

SGSN (Serving GPRS Support Node):

- Delivers packets to/from mobile stations in its service area.
- Handles packet routing, mobility management (attach/detach, location), authentication, and charging.
- Stores location info and user profiles (IMSI, PDP addresses) for registered GPRS users.

GGSN (Gateway GPRS Support Node):

- Gateway between GPRS backbone and external packet data networks (Internet, X.25).
- Converts GPRS packets from SGSN into IP/X.25 format for external networks and vice versa.
- Stores current SGSN address of each user for routing; performs authentication and charging.

📝 **Exam Tip:** Memory: SGSN = Serving (serves the mobile station directly). GGSN = Gateway (gateway to the Internet).

4.3 GPRS Mobility Management — Three MS States

Q13 [Exam]: Explain the GPRS MS three states (mobility management).

✅ **Answer — Q3d context**

State	Description	Location Tracking
IDLE	MS not registered to GPRS. Not reachable. No GPRS attach performed.	Unknown — no tracking at all
READY	MS performed GPRS Attach and is active or recently transmitted. READY timer running.	Full cell-level — every cell change reported to SGSN
STANDBY	READY timer expired. MS registered but inactive.	Routing Area (RA) level — only reports when moving to new RA (several cells)

Transitions: IDLE→READY: GPRS Attach. READY→STANDBY: READY timer expires. STANDBY→READY: packet transmitted. Any→IDLE: GPRS Detach.

4.4 GPRS Quality of Service (QoS)

Q14 [Exam]: Briefly explain four parameters used to define QoS in GPRS.

✅ **Answer — Q3d — 4 marks**

QoS Parameter	Definition	Classes	Real Example
Service Precedence	Importance of data flow — higher priority forwarded first during congestion.	High, Normal, Low	Ambulance alert = High; background app update = Low.
Reliability	Maximum acceptable packet loss, duplication, missequencing, and corruption.	Class 1 (strictest) to Class 3	Bank transaction = Class 1. Weather widget = Class 3.
Delay	Maximum end-to-end transfer time between GPRS interfaces.	Class 1 (<0.5s), Class 2 (<5s), Class 3 (<50s), Best Effort	Vehicle tracking = Class 1. Bulk email = Class 3.
Throughput	Guaranteed peak (maximum) and mean (average) bit rates for the session.	Peak: 8–2048 kbps	Video streaming needs guaranteed mean ≥128 kbps.

MODULE 5

UMTS (3G) & 4G LTE

W-CDMA | UTRAN | QoS Classes | 4G Architecture | EPC | Applications

5.1 UMTS — Universal Mobile Telecommunications System (3G)

UMTS is the 3G successor to GSM, using W-CDMA (Wideband CDMA) radio access. Reuses GSM/GPRS core network components. Base UMTS: 384 kbps downlink; HSDPA variants: 3.6+ Mbps.

UMTS Three Main Components

a) User Equipment (UE): Contains ME + USIM (UMTS Subscriber Identity Module — stores IMSI, authentication algorithms, encryption keys).

b) UTRAN (UMTS Terrestrial Radio Access Network):

- Node B (Base Stations): Converts data between Uu radio interface and Iub. Handles channel coding, interleaving, modulation.
- RNC (Radio Network Controller): Controls radio resources, handles handover, connects to MSC/VLR (circuit) and SGSN (packet).

c) Core Network (CN) — shared with GSM/GPRS: HLR, MSC/VLR, Gateway MSC, SGSN, GGSN.

Q15: Describe the four UMTS QoS traffic classes.

✅ **Answer**

Class	Delay Tolerance	Fundamental Characteristic	Example Applications
Conversational	Strictest — lowest delay	Strict, low delay. Preserves timing between stream entities.	Voice calls, Video telephony, Video games
Streaming	High — less strict	Preserves timing between information entities.	Streaming multimedia, Video on demand
Interactive	Medium	Request-response pattern. Preserves data integrity.	Web browsing, Network games
Background	Least strict	Data integrity preserved. No strict delay requirement.	Background email download, File transfers

📝 **Exam Tip:** Memorize in order: Conversational → Streaming → Interactive → Background. Conversational = most delay-sensitive; Background = least.

5.2 4G LTE — Long Term Evolution

4G LTE is a system architecture REVOLUTION — a completely new all-IP (EPS: Evolved Packet System) design. Key innovation: Control Plane (signaling) and User Plane (data) handled by SEPARATE network elements.

Term	Definition
eNodeB	4G base station. Works INDEPENDENTLY — no central controller needed (unlike 3G's RNC). Peer connections with other eNodeBs via X2 interface.
MME	Mobility Management Entity — handles SIGNALING: authentication, authorization, session setup. Does NOT carry user data.
SGW	Serving Gateway — routes and forwards USER DATA packets. Handles inter-eNodeB handover to eliminate data gaps.
PGW	PDN Gateway — connects to external networks (Internet). Assigns IP addresses, applies QoS.
HSS	Home Subscriber Server — replaces HLR in 4G. Stores all subscriber data.

Memory: MME = Management (signaling). SGW/PGW = data highways.

Applications of 4G

- Data-intensive interactive services — online satellite mapping loads instantly.
- Multiple-user video conferencing — see and talk to more than one person.
- Location-based services — real-time weather/traffic, find nearby businesses.
- Tele-medicine — monitor/advise remote patients with live video streaming.
- HDTV — high-definition TV channels streamed directly to subscriber devices.

MODULE 6

5G — Fifth Generation Mobile Networks

eMBB | URLLC | mMTC | CUPS | UPF | AMF | MEC | Network Slicing | Beamforming | Massive MIMO

6.1 Key 5G Specifications

Feature	5G	4G (Comparison)
Peak Speed	Up to 20 Gbps	~150 Mbps — 5G is ~100x faster
Latency	As low as 1 millisecond	30–50ms — 5G is 50x lower
Device Density	1 million devices per km²	~4,000 per km² — 5G supports 250x more
Energy Efficiency	90% less energy per bit	Far less efficient
Reliability	99.999% uptime	Critical for life-saving services

6.2 Three 5G Service Types — Most Important for Exams

Q16: True or False: Network Slicing allows a single physical 5G network to be partitioned into multiple virtual networks with different performance characteristics.

✅ Answer — Answer: TRUE □

This statement is TRUE. Network Slicing creates multiple independent virtual networks on ONE shared physical 5G infrastructure. Each slice has independently configured bandwidth, latency, and reliability — completely isolated from each other.

Example: One 5G network sliced into: Emergency/ambulance slice (guaranteed lowest latency), Consumer streaming slice (high bandwidth), IoT sensor slice (massive connections). All run on the same physical towers.

Service	Full Name	Priority	Applications	Real Example
eMBB	Enhanced Mobile Broadband	Maximum SPEED	4K/8K video, AR/VR, cloud gaming, fast downloads	80,000 fans streaming a stadium match in HD simultaneously — 5G eMBB handles all.
URLLC	Ultra-Reliable	Minimum DELAY	Self-driving cars,	Surgeon in

	Low-Latency Communication	+ Maximum RELIABILITY	remote surgery, industrial robots	London controls robotic arm in Nairobi — delay over 5ms could cost patient's life. 5G URLLC makes this safe.
mMTC	Massive Machine-Type Communications	Maximum NUMBER OF DEVICES	Smart meters, agriculture sensors, smart city	Smart city with 500,000 sensors per km² monitoring traffic, air, water — all via 5G mMTC.

📝 **Exam Tip:** Memory: eMBB = SPEED (your phone). URLLC = LIFE-CRITICAL (surgery/cars). mMTC = BILLIONS of small IoT devices.

6.3 Key 5G Technologies

Q17: True or False: Massive MIMO involves using a very small number of antennas to save space on cell towers.

✅ **Answer — Answer: FALSE □**

FALSE. Massive MIMO uses 64–256+ antennas at a single base station. 'Massive' = massively LARGE number of antennas. This allows the base station to serve many users simultaneously using spatial multiplexing, dramatically increasing spectral efficiency.

Q18: True or False: Beamforming broadcasts a signal in all directions equally to ensure maximum coverage.

✅ **Answer — Answer: FALSE ✗**

FALSE. Beamforming FOCUSES the signal like a SPOTLIGHT directly at a specific user. It is the OPPOSITE of omnidirectional broadcasting. By controlling phase and amplitude from multiple antennas, signal energy is concentrated toward the intended receiver — not wasted in all directions.

Analogy: Traditional antenna = light bulb (illuminates room equally). Beamforming = spotlight (follows you specifically).

Technology	Simple Explanation	Real Example
mmWave	Very high frequency (24–100 GHz). Extremely fast but short range (~100–200m). Blocked by walls.	5G mmWave in a stadium — antennas installed every 100m throughout venue.
Sub-6 GHz	Below 6 GHz. Good range,	Sub-6 GHz 5G covering rural

	penetrates walls. Wide area coverage.	Tanzania — better speed over large distances.
Massive MIMO	64–256+ antennas at base station simultaneously serve many users.	5G tower in Dar es Salaam CBD serving hundreds of users simultaneously.
Beamforming	Focuses signal like a spotlight at each specific user.	Signal always directed at you — not wasted in empty space around you.
Network Slicing	One physical 5G network → multiple isolated virtual networks, each for a different purpose.	Emergency slice + consumer slice + IoT slice — all on same towers, fully isolated.
Edge Computing (MEC)	Processing at base station near user — not in distant cloud.	Self-driving car: collision detection at local edge server (1ms) not distant cloud (50ms).

6.4 5G Core Architecture — CUPS, UPF, AMF

Q19: True or False: Control and User Plane Separation (CUPS) allows the control logic to scale independently from data traffic.

✅ Answer — Answer: TRUE ✓

TRUE. CUPS is a fundamental 5G principle that separates: CONTROL PLANE (signaling, authentication, session management) from USER PLANE (actual data forwarding). Each scales independently. This enables MEC — the User Plane can be deployed at the network edge near the user while control remains centralized.

5G Function	Handles Signaling?	Handles Data Traffic?	Analogy
AMF (Access & Mobility Management)	YES — authentication, registration, mobility	NO	Airport immigration — checks identity and grants entry.
UPF (User Plane Function)	NO	YES — forwards all user data packets to/from internet	The highway — carries all actual traffic.
SMF (Session Management)	YES — sets up sessions, assigns IPs	NO	Booking clerk — configures the session and highway.
AUSF (Auth Server)	YES — authentication	NO	Security database verifying identity.

Q20: True or False: The User Plane Function (UPF) in 5G handles the signaling

and authentication of users.

✅ **Answer — Answer: FALSE ✗**

FALSE. UPF handles actual USER DATA TRAFFIC (packets between UE and the internet). Signaling and authentication are handled by AMF (and AUSF for authentication specifically). 'User Plane' = data traffic, NOT signaling.

6.5 Mobile Edge Computing (MEC)

Q21: Which 5G feature is most critical for ultra-low latency in MEC? (A) Massive MIMO (B) Network Slicing (C) CUPS (D) Beamforming

✅ **Answer — Test Paper 2 Q iv — Answer: C**

CUPS is the most critical feature for MEC latency. It allows the USER PLANE (data traffic) to be deployed at the network EDGE — physically close to the user — while control remains centralized. Data travels a very short path to the nearby edge server: ~1ms instead of 50–100ms to a distant cloud.

Q22: In a MEC environment, where does AI inferencing typically occur? (A) Centralized Cloud (B) Edge Node/Server (C) UE only (D) Core Network

✅ **Answer — Test Paper 2 Q v — Answer: B**

AI inferencing runs at the EDGE NODE/SERVER — computing hardware at or near the base station. Example: A self-driving car needs to detect a pedestrian in 1ms. Inferencing at nearby edge server = 1ms. At distant cloud = 50ms round trip. Only MEC makes real-time AI decisions physically safe.

Q23: What is the primary driver for moving compute to the edge in MEC? (A) Increase device storage (B) Reduce latency and congestion (C) Eliminate base stations (D) Provide free internet

✅ **Answer — Test Paper 2 Q vi — Answer: B**

PRIMARY driver: Reduce network congestion AND round-trip latency. By processing near the user: (1) Round-trip drops from 50–100ms to ~1ms. (2) Network congestion reduced — only results travel wider network, not raw data. Essential for autonomous vehicles, remote surgery, and industrial robots.

MODULE 7

Cloud Computing

5 Characteristics | IaaS / PaaS / SaaS | Deployment Models | Serverless / FaaS

7.1 What is Cloud Computing?

Cloud computing is the delivery of computing services — servers, storage, databases, software — over the internet. Instead of buying hardware, you rent computing power and pay ONLY for what you use.

Analogy: Electricity. You don't build your own power station — you plug into the national grid and pay per unit. Cloud computing is the same concept for computing power.

7.2 Five Essential Characteristics (NIST Definition)

Characteristic	Meaning	Real Example
On-Demand Self-Service	Get resources automatically — no human from provider needed.	Startup provisions 10 servers via AWS console in 5 minutes.
Broad Network Access	Access from anywhere — phone, laptop, tablet.	Student in Mwanza accesses Dar es Salaam university library from phone.
Resource Pooling	Provider serves many customers on shared hardware (data isolated).	AWS serves millions of businesses from shared data centers.
Rapid Elasticity	Scale up or down instantly.	Amazon adds thousands of servers at midnight on Black Friday, removes them by morning.
Measured Service	Pay only for what you use.	Company pays for exactly 500GB used this month — no fixed cost.

7.3 Cloud Service Models

Q24: Which term is used interchangeably with Serverless computing? (A) IaaS (B) PaaS (C) FaaS (D) SaaS

✅ **Answer — Test Paper 2 Q vii — Answer: C — Function as a Service**

Serverless = FaaS (Function as a Service). Code runs as individual FUNCTIONS triggered by events. Provider handles all server management. Developer writes only the function. Pay only for

actual execution time (milliseconds). AWS Lambda, Google Cloud Functions, Azure Functions are examples.

Model	Full Name	You Manage	Provider Manages	Best For	Real Example
IaaS	Infrastructure as a Service	OS, software, apps	Hardware, virtualisation, networking	IT companies wanting full control.	Startup rents Amazon EC2 virtual server for KES 2,000/month instead of buying hardware for KES 200,000.
PaaS	Platform as a Service	Application code only	Everything underneath	Developers.	Developer builds mobile banking app on Heroku — never touches a server.
SaaS	Software as a Service	NOTHING	Everything	Everyday users.	500 employees use Microsoft 365 — IT never manages updates; employees always have latest version.
FaaS / Serverless	Function as a Service	Function code only	All servers + scaling + triggers	Event-driven applications.	AWS Lambda resizes uploaded images — runs only when photo uploaded, charges only for milliseconds of execution.

📝 **Exam Tip:** Exam analogy: IaaS = empty apartment (you furnish it). PaaS = furnished apartment (just move in). SaaS = hotel room (everything done for you). This analogy always earns marks.

7.4 Cloud Deployment Models

Q25: What does a deployment model primarily describe in cloud computing?

✅ Answer — Test Paper 2 Q i — Answer: B. The location and management of the cloud's infrastructure

A deployment model describes WHERE the cloud infrastructure physically sits, WHO owns it, and WHO controls and accesses it. It does NOT describe programming languages, billing methods, or what is being built — it describes the CONTROL and ACCESS structure.

Q26: An organization uses on-premises infrastructure for daily operations but bursts into a public cloud during peak traffic. This is: (A) Private (B) Public (C) Hybrid (D) Community Cloud

✅ Answer — Test Paper 2 Q ii — Answer: C — Hybrid Cloud

HYBRID CLOUD — uses BOTH private (on-premises for normal operations) AND public cloud (for peak traffic bursting). The 'bursting' pattern is the classic Hybrid Cloud use case. Private + Public working together = Hybrid.

Q27: Primary disadvantage of Private Cloud vs Public Cloud? (A) Lower security (B) Limited customization (C) Higher upfront capital expenditure (D) Shared hardware

✅ Answer — Test Paper 2 Q iii — Answer: C

PRIMARY disadvantage = HIGHER UPFRONT CAPITAL EXPENDITURE. Organizations must purchase, install, and maintain their own hardware — large capital cost before any benefit. Public Cloud = zero upfront, pay-as-you-go.

Why others are wrong: Private Cloud has BETTER security (not lower). Private Cloud has MORE customization (not limited). Private Cloud does NOT share hardware — that describes Public Cloud.

Deployment Model	Who Owns/Controls	Who Can Access	Real Example
Private Cloud	Single organization — full control.	Only that organization.	Hospital running patient data on dedicated private servers.
Public Cloud	Third-party provider (AWS/Azure/Google).	Anyone who subscribes.	Startup using AWS — shares physical servers with thousands of businesses.
Hybrid Cloud	Both — organization + provider.	Depends on component.	Company uses own servers normally; bursts into AWS during

			Black Friday peak traffic.
Community Cloud	Shared by community or third party.	Only specific community members.	Multiple government agencies sharing a compliance-ready cloud.

MODULE 8

Internet of Things (IoT)

Architecture Layers | How It Works | Applications | Security | AI in Mobile Computing

8.1 What is IoT?

IoT is the network of physical objects — machines, appliances, sensors, vehicles — embedded with sensors and software that connect to the internet and automatically collect and exchange data WITHOUT human involvement.

Formula: IoT = Smart Object + Internet Connection + Automatic Data Collection + Automatic Action.

8.2 Standard IoT Three-Layer Architecture

Q28: In the standard IoT architecture, which layer is responsible for sensing and gathering data? (A) Application (B) Network (C) Perception (D) Support Layer

✅ **Answer — Test Paper 2 Q viii — Answer: C — Perception Layer**

The PERCEPTION LAYER (also called Sensing Layer or Device Layer) senses and gathers data from the physical world. It contains all physical IoT devices: sensors, actuators, cameras, RFID readers, GPS trackers. They convert physical conditions into digital data.

Memory: Perception = what you PERCEIVE (see, feel, detect) from the physical world.

Layer	Also Called	Responsibility	Components	Real Example
Perception Layer	Sensing Layer / Device Layer	SENSING — the 'eyes and ears' of IoT.	Sensors (temp, moisture, motion, RFID), actuators, cameras.	Soil moisture sensor detecting 20% water level in a farm field.
Network Layer	Transport Layer	TRANSMITTING data to processing systems — the 'nervous system'.	WiFi, 4G/5G, LoRaWAN, Bluetooth, ZigBee, NB-IoT.	Sending moisture reading via LoRaWAN to cloud server 10km away.
Application Layer	Service Layer	PROCESSING data, delivering intelligent services — the 'brain'.	Cloud platforms, AI analytics, dashboards, automation, user apps.	Cloud decides irrigation needed → switches it on → alerts farmer on phone.

8.3 How IoT Works — 4 Stages

Stage	What Happens	Real Example
1 — Sense	Sensor detects and measures physical condition.	Soil moisture sensor detects moisture dropped to 20% — too dry for crops.
2 — Connect	Measured data transmitted wirelessly to server.	Sensor sends reading via LoRaWAN to cloud server 10km away.
3 — Process	Cloud/edge computing analyzes data.	Cloud: moisture low + hot forecast = irrigation needed now.
4 — Act	Automatic decision and action executed.	Irrigation system on for 20 minutes. Farmer receives phone alert.

8.4 IoT Security Challenge

Q29: Which is a significant security risk for IoT? (A) Too many passwords (B) Hardcoded default passwords and lack of encryption (C) Devices too fast to track (D) Too much recycled plastic

✅ Answer — Test Paper 2 Q ix — Answer: B

PRIMARY IoT security risk: HARDCODED DEFAULT PASSWORDS + LACK OF ENCRYPTION. Most IoT devices ship with simple credentials (admin/admin) users never change. Without encryption, transmitted data is visible in plain text. Together: millions of easily exploitable entry points.

Real Evidence — Mirai Botnet (2016): Hackers found 600,000 IoT devices with default passwords, infected them all, launched the largest DDoS in history — took down Twitter, Netflix, Amazon for hours. Caused entirely by unsecured IoT devices.

8.5 IoT Applications

Area	How IoT Helps	Real Example
Healthcare	Continuous remote patient monitoring.	Apple Watch detected atrial fibrillation — alerted user to see cardiologist. Early detection potentially saved their life.
Smart City	Thousands of sensors manage traffic, waste, parking, air quality.	Barcelona IoT bins signal when 80% full → trucks only visit full bins → 40% route reduction, 37% fuel savings.
Agriculture	Sensors, drones, automated	California vineyards: 1,000+ soil

	systems for precision farming.	sensors per farm → 25% less water used, improved grape quality.
Smart Home	Automated home tasks based on sensors.	Google Nest thermostat learns schedule → starts warming at 5:45pm, stops heating automatically when you go on holiday.

8.6 AI Applications in Mobile Computing

Q30: A Chatbot is an application of which AI technology? (A) Robotics (B) NLP (C) Computer Vision (D) Expert Systems

✅ Answer — Test Paper 2 Q x — Answer: B — Natural Language Processing

A Chatbot is powered by NLP (Natural Language Processing) — AI that allows computers to understand, interpret, and respond to HUMAN LANGUAGE conversationally. The chatbot processes the user's text/speech input (NLP task) and generates an appropriate response.

Why others are wrong: Robotics = physical mechanical systems. Computer Vision = images/video (not text). Expert Systems = rule-based knowledge bases for specific domains.

AI Technology	Definition	Mobile Application	Real Example
NLP (Natural Language Processing)	AI understanding human language (text and speech).	Chatbots, voice assistants, auto-translation.	Banking chatbot understanding 'What is my balance?'. Google Translate on mobile.
Computer Vision	AI interpreting visual information from images/video.	Facial recognition, object detection, AR.	Face unlock on smartphones. Self-driving car cameras. Medical image scanning apps.
Robotics	AI combined with mechanical systems.	Industrial automation, drone delivery.	Amazon warehouse robots. Drone delivery of medicines to remote areas.
Expert Systems	Rule-based AI mimicking a domain expert.	Medical diagnosis, fault diagnosis, financial advisory.	Mobile medical app asking about symptoms and recommending treatment.
Machine Learning	AI learning from data to improve predictions.	Recommendation systems, fraud detection, predictive text.	Spotify song recommendations. Auto-complete on keyboard.

MODULE 9

Mobile IP

Home Agent | Care-of Address | Tunneling | Triangular Routing | Route Optimization

9.1 Why Mobile IP Is Needed

The internet was designed for STATIONARY devices — IP addresses are tied to network locations. When a device moves to a new network, it gets a new IP address and all active connections break. Mobile IP solves this: the device keeps its permanent home IP and connections continue seamlessly.

Without Mobile IP: Moving to a new network breaks ALL connections — calls drop, downloads fail, streaming stops.

With Mobile IP: Home Agent forwards traffic to new location — connections continue unnoticed.

Term	Definition
MN (Mobile Node)	Your device — has a permanent home IP that NEVER changes. Moves between networks.
HA (Home Agent)	Router in your home network. Intercepts incoming traffic for MN, tunnels it to current COA.
FA (Foreign Agent)	Router in the visited foreign network. Receives tunneled packets from HA and delivers to MN.
COA (Care-of Address)	Temporary IP representing MN's CURRENT LOCATION in the foreign network.
Tunneling	HA wraps original packet inside a new packet addressed to COA — IP-in-IP encapsulation.
Binding	MN's registration of current COA with HA so HA can correctly tunnel incoming traffic.

Q31 [Exam]: What is a Care-of Address (COA)? Briefly explain its two types.

✅ **Answer — CBE Q4a — 6 marks**

COA is a temporary IP assigned to a Mobile Node when visiting a foreign network — representing its CURRENT LOCATION. The Home Agent uses the COA to tunnel data. COA is NEVER stored in DNS — other nodes always look up the permanent home address; HA handles forwarding transparently.

Type	Description	How Obtained	Key Feature

Foreign Agent COA (FA-COA)	FA router's IP used as COA. FA receives all tunneled packets from HA and delivers to MN.	From FA's periodic Agent Advertisement messages.	SHARED — multiple MNs use same FA-COA. Saves IP addresses.
Co-located COA (CCoA)	Unique temporary IP assigned DIRECTLY to MN's own network interface.	Via DHCP from foreign network.	UNIQUE per MN. Works even with no FA. Enables route optimization more easily.

Q32 [Exam]: Briefly describe THREE properties of Care-of Address.

✅ Answer — CBE Q4b — 3 marks

- Foreign-link specific: Valid only within the particular foreign network currently visited. Becomes invalid as soon as MN moves to a different network.
- Dynamically changing: Generally changes every time MN moves to a new foreign network. HA must always have the most current COA to forward traffic correctly.
- Exit-point of HA tunnel: COA is the TERMINATION POINT of the tunnel created by HA. HA encapsulates incoming packets (addressed to permanent home IP) in a new tunnel packet whose outer destination header = COA.

Q33 [Exam]: Explain triangular routing problems in Mobile IP and the solution.

✅ Answer — CBE Q4c — 8 marks

Triangular Routing: CN → Home Agent (HA) → Foreign Agent (FA) → Mobile Node (MN). Data always routes through HA first — even when CN and MN may be geographically close.

Six Problems with Triangular Routing:

- Sub-optimal routing — data takes a long path even when CN and MN are nearby.
- Increased latency — extra HA hops degrade real-time voice/video calls.
- HA bottleneck — all roaming traffic through HA = single point of congestion and failure.
- Firewall problem (Ingress Filtering) — MN's outgoing data with home IP source may be rejected by foreign network's firewall.
- Multicast problem — MN loses group membership if it sends directly from foreign network.
- TTL problem — packet may expire before completing the longer triangular path.

Solution 1 — Reverse Tunneling: MN → FA → HA → CN for OUTGOING traffic. Fixes ingress filtering, multicast, and TTL issues.

Solution 2 — Route Optimization (RFC 3775): MN sends Binding Update directly to CN. CN stores COA in Binding Cache, sends future data DIRECTLY to COA: CN → FA → MN. Completely bypasses HA.

Q34 [Exam]: Differentiate between Mobility and Nomadicity.

✅ **Answer — CBE Q4d — 3 marks**

Aspect	Mobility	Nomadicity
Definition	Capability to physically move between locations/networks while maintaining connectivity.	Computing paradigm where the entire environment (apps, data, preferences) travels WITH the user — accessible from any location or device.
Focus	The ACT of movement.	CONTINUITY of computing experience across locations and devices.
Technical need	Wireless networks, Mobile IP, handover mechanisms.	Cloud storage, VPNs, sync tools, device-independence.
Example	Walking on a 4G video call — towers switch automatically.	Professor carrying laptop with all notes and grades everywhere.

MODULE 10

TCP/IP in Mobile Networks

4-Layer Model | TCP vs UDP | TCP Problems in Wireless | Solutions

10.1 TCP/IP Model vs OSI

TCP/IP is the 4-layer practical model that the internet actually uses. OSI is the 7-layer theoretical model for understanding. TCP/IP maps as: Application (OSI 5-7), Transport (OSI 4), Internet (OSI 3), Network Interface (OSI 1-2).

10.2 TCP Problems in Wireless Networks

TCP was designed in 1974 for WIRED networks. It assumes packet loss = network congestion. In wireless networks, packet loss is caused by SIGNAL FADING — not congestion. This mismatch causes major problems:

TCP Problem in Wireless	Description	Impact
False congestion detection	TCP mistakes wireless signal loss for congestion → reduces transmission rate unnecessarily.	Throughput drops even when network has capacity.
Slow start after loss	TCP enters slow-start mode after any packet loss — takes a long time to recover speed.	Long recovery periods from brief fades.
High BER interaction	Wireless BER (10^{-3}) causes frequent retransmissions → ACK storms.	Wastes bandwidth and battery.
Handover disruption	Moving between cells causes brief disconnection → TCP times out → treats as congestion.	Speed drops dramatically after every handover.

10.3 Solutions for TCP in Wireless

- Indirect TCP (I-TCP): Split TCP connection at base station — wireless segment handled separately with optimized parameters.
- Snoop TCP: Agent at base station sniffs ACKs, locally retransmits lost packets without informing sender — prevents false congestion signals.
- Mobile TCP (M-TCP): Splits connection. If wireless link fails, suspends sender without reducing window size — quick resume when link recovers.
- QUIC Protocol: Google's modern UDP-based transport with reliability built in — better suited for mobile than TCP.

MODULE 11

Mobile Ad Hoc Networks — MANET

Self-organizing | Multi-hop | Routing Protocols | Proactive vs Reactive | Constraints | Applications

11.1 What is a MANET?

A MANET (Mobile Ad Hoc Network) is a self-organizing, infrastructure-less wireless network where mobile nodes communicate directly peer-to-peer without any fixed base stations or access points. Every device is BOTH a user AND a router — forwarding data for other devices.

Q35 [Exam]: How does MANET differ from WLAN?

✅ **Answer — CBE Q5a — 4 marks**

Feature	WLAN (Wireless LAN)	MANET
Infrastructure	REQUIRED — access points, routers, wired backbone installed before use.	NONE — devices connect directly with zero pre-installed hardware.
Device role	END USERS only — communicate through AP.	BOTH user AND router — each forwards data for others (multi-hop).
Setup time	Hours to days — IT admin configures.	Seconds to minutes — automatic self-configuration.
Centralized control	YES — AP controls all transmissions.	NO — fully decentralized.
Failure point	AP failure = entire local network down.	Distributed — auto re-routing around failures.
Best for	Permanent locations: offices, homes, campuses.	Temporary/emergency environments without infrastructure.

11.2 MANET Routing Protocols

Q36: True or False: Reactive routing protocols have higher control overhead than proactive protocols because they update tables constantly.

✅ **Answer — Answer: FALSE ✗**

FALSE. It is PROACTIVE protocols that have higher control overhead — they continuously update routing tables to ALL destinations whether or not data is being sent. REACTIVE protocols generate routing traffic ONLY when data needs to be sent → lower overhead.

Q37: True or False: AODV is an example of a proactive routing protocol.

✅ **Answer — Answer: FALSE ✗**

FALSE. AODV (Ad Hoc On-Demand Distance Vector) is REACTIVE (on-demand). The name itself reveals this — 'On-demand' means routes discovered ONLY when data needs to be sent. Proactive examples: DSDV, OLSR.

Q38: True or False: Zone Routing Protocol (ZRP) uses a proactive approach for nearby nodes and a reactive approach for distant nodes.

✅ **Answer — Answer: TRUE ✓**

TRUE. ZRP is a HYBRID protocol — proactive within the local zone (routes maintained for fast communication with nearby nodes), reactive outside the zone (routes discovered on demand to save bandwidth for far destinations). Best of both approaches.

Feature	Proactive (Table-Driven)	Reactive (On-Demand)	Hybrid (ZRP)
Route building	Maintained CONTINUOUSLY to all destinations.	Built ONLY when data needs to be sent.	Proactive locally, reactive globally.
Control overhead	HIGH — constant updates even in silence.	LOW — zero routing traffic when idle.	MEDIUM.
Route discovery delay	ZERO — route already known.	Small delay — must discover route first.	Zero locally; small delay globally.
Best for	Small, slow-moving, high-traffic networks.	Large networks with sporadic traffic.	Large networks needing both speed and efficiency.
Examples	DSDV, OLSR	AODV, DSR	ZRP

11.3 MANET Unicast vs Broadcast vs Multicast

Q39: True or False: Unicast routing in a MANET refers to sending a message from one source to ALL other nodes.

✅ **Answer — Answer: FALSE ✗**

FALSE. Unicast = ONE source to ONE specific destination. Sending to ALL nodes = BROADCAST. Sending to a defined GROUP = MULTICAST.

Type	Definition	MANET Example

Unicast	One source → ONE specific destination.	Soldier sends command to a specific team leader.
Broadcast	One source → ALL nodes in network.	Commander broadcasts emergency alert to every soldier's device.
Multicast	One source → SPECIFIC GROUP of nodes.	Commander sends mission details to only the 5 members of one squad.
Geocast	One source → all nodes in a GEOGRAPHIC AREA.	Warning sent to all vehicles within 500m radius of an accident.

11.4 MANET Applications

Q40 [Exam]: Briefly describe THREE applications of Mobile Ad-Hoc Networks.

✅ **Answer — CBE Q5b — 3 marks**

1. Tactical Military Networks:

Soldiers in remote terrain need communication without fixed infrastructure that could be targeted. Each soldier's radio connects to nearby devices forming a multi-hop network. Example: Messages hop soldier-to-soldier to reach the commander. If a device is destroyed, network auto re-routes.

2. Emergency Services and Disaster Recovery:

Natural disasters destroy all communication infrastructure. Rescue teams arrive with MANET devices and have a working network within minutes. Example: 2010 Haiti earthquake destroyed all towers — international rescue teams used MANET radios for immediate coordination.

3. Vehicle-to-Vehicle (V2V) Communication:

Vehicles share real-time safety information (hazards, road conditions) without infrastructure for every message. Example: Car detects black ice → alert hops through vehicles behind via MANET in milliseconds — drivers far back receive warning before reaching the hazard.

11.5 MANET Constraints

Q41 [Exam]: Explain any FOUR constraints facing Mobile Ad-Hoc Networks.

✅ **Answer — CBE Q5c — 8 marks**

1. Infrastructure-less — New Network Design Challenges:

No routers, APs, DHCP, or DNS. All networking functions must be distributed across devices. Every device simultaneously participates in routing, addressing, and organization. Protocols must be designed from scratch for a fully decentralized environment.

2. Dynamically Changing Topologies:

Devices move constantly — network map changes every few seconds. Consequences: route breakage (transfers interrupted mid-flow), network partitions (isolated segments), packet loss. Example: Rescue worker walking into a building breaks the MANET route through them — all traffic on that path is lost instantly.

3. Physical Layer Limitations:

- Limited range — direct communication only ~100–300m; longer requires multi-hop.
- High packet loss — radio interference, walls, distance cause far higher error rates than wired.
- Broadcast nature — all nearby devices receive all transmissions → interference and privacy risks.

4. Limited Node Resources:

- Battery drain — acting as routing hub for 20 devices drains battery far faster than just being an end user.
- Limited CPU — complex routing algorithm calculations difficult on low-power embedded devices.
- Limited memory — routing tables for hundreds of nodes may exceed what resource-limited devices have.

11.6 Bandwidth-Efficient Routing — AODV

Q42 [Exam]: For effective bandwidth utilization, which routing protocol is best for MANETs? Justify.

✅ **Answer — CBE Q5d — 5 marks**

Recommendation: AODV (Ad Hoc On-Demand Distance Vector) — a REACTIVE protocol.

Justification: Wireless channel in MANET is limited and shared. Proactive protocols (e.g., DSDV) waste this bandwidth with continuous routing updates to ALL neighbours — even during complete silence.

Why AODV is better:

- Routes discovered ONLY when data needs to be sent — zero routing overhead during silence.
- Entire bandwidth preserved for actual data during idle periods.
- Scales efficiently — 500-node network doesn't flood itself when only 5 nodes communicate.

AODV Operation: Source broadcasts RREQ (Route Request) → nodes rebroadcast if no route → Destination sends RREP (Route Reply) back → Source uses discovered route → RERR (Route Error) if link breaks → rediscover.

Conclusion: AODV is best because it generates routing traffic ONLY when needed, preserving scarce wireless bandwidth for actual data — critical in battery-powered, bandwidth-limited environments.

EXAM SECTION

Past Paper — Full Solved

College of Business Education | ITU07401 | July 2018 | 3 Hours | 100 Marks | 5 Questions × 20 Marks

Green boxes = actual exam questions exactly as set. Answer follows immediately below each.

QUESTION ONE — 20 Marks

Q1a [Exam]: Q1(a) — 2 marks: Define the term Mobile Computing.

✅ **Answer**

Mobile Computing is a technology that allows transmission of data, voice, and video via a computer or any other wireless-enabled device without being connected to a fixed physical link. Example: A doctor in a moving ambulance accessing patient records on a tablet.

Q1b [Exam]: Q1(b) — 9 marks: Briefly explain FOUR characteristics and TWO constraints of mobile computing.

✅ **Answer — See Module 2 Q3 above for full answer with examples**

FOUR CHARACTERISTICS: Mobility, Portability, Wireless Communication, Connectivity. TWO CONSTRAINTS: Limited Resources (battery/CPU/memory), Variable Connectivity (signal changes with location/obstacles). Each characteristic/constraint = ~1.5 marks. Always include a real-world example for full marks.

Q1ci [Exam]: Q1(c)(i) — 5 marks: Differentiate between User Mobility and Host Mobility.

✅ **Answer — See Module 2 keyTable above**

User Mobility = PERSON moves and continues same service using same login credentials. IP may change — service follows identity. Host Mobility = DEVICE moves between networks while maintaining a reachable IP address using Mobile IP mechanisms (Home Agent, COA, tunneling). Requires Mobile IP protocols.

Q1cii [Exam]: Q1(c)(ii) — 4 marks: Differentiate between TDMA and FDMA.

✅ **Answer — See Module 2 Q7 comparison table above**

FDMA divides BANDWIDTH into frequency channels — each user gets own dedicated frequency, all transmit simultaneously. TDMA divides TIME into slots — all share one frequency, taking turns.

GSM uses BOTH: FDMA splits spectrum into 200 kHz channels, TDMA splits each into 8 time slots.

QUESTION TWO — 20 Marks

Q2a [Exam]: Q2(a) — 8 marks: With illustrations, explain how Exposed Terminal differs from Hidden Terminal Problem.

✅ **Answer — See Module 2 Q4 for full step-by-step answer with setups**

Hidden Terminal: A──►B◄──C. A and C cannot hear each other. C falsely senses channel as free and transmits → COLLISION at B. Reduces throughput. Solution: RTS/CTS. Exposed Terminal: A◄──B────►C──►D. C hears B, wrongly waits. C's transmission to D would not interfere. Wastes bandwidth. Key: Hidden = collision occurs. Exposed = no collision, wasted opportunity.

Q2b [Exam]: Q2(b) — 6 marks: Explain how Delay and Throughput compare MAC protocols.

✅ **Answer — See Module 2 Q6 above for full answer**

Delay = average time from packet queuing to complete transmission (includes queuing, transmission, propagation, retransmission delays). Lower = better for real-time apps. Throughput $\eta = P/(T \times C)$ = fraction of channel capacity used for useful data. These metrics TRADE OFF — reducing delay aggressively increases collisions and reduces throughput.

Q2c [Exam]: Q2(c) — 6 marks: How are RTS and CTS used to avoid collision?

✅ **Answer — See Module 2 Q5 above for full step-by-step answer**

RTS (Request to Send) and CTS (Clear to Send) reserve the wireless channel before data transmission. Sender sends RTS → Receiver replies with CTS broadcast to ALL nearby nodes → All nodes hearing CTS set NAV timer and stay silent → Sender transmits data → ACK confirms receipt. Solves hidden terminal: hidden node C hears CTS from B → stays silent even though it never heard A's RTS.

QUESTION THREE — 20 Marks

Q3a [Exam]: Q3(a) — 2 marks: What is GSM? In which frequency bands?

✅ **Answer — See Module 3 Q9 with frequency table**

GSM = Global System for Mobile Communication, a 2G digital standard using TDMA+FDMA. Bands: GSM-900 (890–960 MHz), GSM-1800 (1710–1880 MHz), GSM-850 (824–894 MHz), GSM-1900 (1850–1990 MHz). East Africa uses GSM-900 and GSM-1800.

Q3b [Exam]: Q3(b) — 4 marks: Differentiate Tele Services and Supplementary Services.

✅ **Answer — See Module 3 Q10 comparison table**

Tele Services = primary end-to-end communication services (voice calls, SMS, emergency calling). Supplementary Services = optional enhancements ON TOP of Tele Services (call forwarding, call hold, call waiting, multiparty). Key: Tele Services work alone; Supplementary Services CANNOT exist without a Tele Service.

Q3c [Exam]: Q3(c) — 10 marks: With GSM architecture, explain the basic parts of NSS.

✅ **Answer — See Module 3 Q11 full answer with component table**

NSS (Network Subsystem) = the 'brain' of GSM. Five components: HLR (permanent subscriber database), MSC (telephone exchange — routes calls), VLR (temporary database for visitors), AUC (authentication: A3, A8, A5 algorithms), EIR (IMEI database — blocks stolen/defective phones). Real example: Traveling from DSM to Mwanza — your HLR stays in DSM, Mwanza MSC creates a VLR copy for fast local service.

Q3d [Exam]: Q3(d) — 4 marks: Briefly explain four QoS parameters in GPRS.

✅ **Answer — See Module 4 Q14 full table**

Four GPRS QoS parameters: (1) Service Precedence — priority (High/Normal/Low) during congestion. (2) Reliability — max acceptable packet loss/corruption (Class 1 strictest to Class 3). (3) Delay — max end-to-end transfer time (Class 1 <0.5s to Best Effort). (4) Throughput — guaranteed peak and mean bit rates.

QUESTION FOUR — 20 Marks

Q4a [Exam]: Q4(a) — 6 marks: What is Care-of Address (COA)? Explain its two types.

✅ **Answer — See Module 9 Q31 full answer with table**

COA = temporary IP address representing the Mobile Node's CURRENT LOCATION in a foreign network. Two types: (1) FA-COA — uses Foreign Agent's IP as COA. SHARED by multiple MNs. Saves IP addresses. (2) Co-located COA (CCoA) — unique temporary IP assigned DIRECTLY to MN's interface via DHCP. UNIQUE per MN. Works without a FA.

Q4b [Exam]: Q4(b) — 3 marks: THREE properties of Care-of Address.

✅ **Answer — See Module 9 Q32**

(1) Foreign-link specific — valid only in the current foreign network. (2) Dynamically changing — changes each time MN moves to a new network; HA must have current COA. (3) Exit-point of HA tunnel — the termination point of the tunnel; COA appears in the outer packet header of tunneled packets.

Q4c [Exam]: Q4(c) — 8 marks: Explain triangular routing problems and the solution.

✅ **Answer — See Module 9 Q33 full answer**

Triangular Routing: CN → HA → FA → MN — data always routes through HA even if CN and MN are geographically close. Six Problems: sub-optimal routing, increased latency, HA bottleneck, firewall/ingress filtering, multicast failure, TTL expiry. Solutions: Reverse Tunneling (MN→FA→HA→CN for outgoing — fixes ingress/multicast/TTL) + Route Optimization RFC 3775 (MN sends Binding Update to CN → CN routes DIRECTLY to COA bypassing HA).

Q4d [Exam]: Q4(d) — 3 marks: Differentiate between Mobility and Nomadicity.

✅ **Answer — See Module 9 Q34 table**

Mobility = capability to physically move between locations while maintaining connectivity (focus on the act of movement — requires Mobile IP, handover). Nomadicity = entire computing environment (apps, data, preferences) travels WITH the user, accessible from any device/location (focus on continuity of experience — requires cloud, sync, VPNs).

QUESTION FIVE — 20 Marks

Q5a [Exam]: Q5(a) — 4 marks: How does MANET differ from WLAN?

✅ **Answer — See Module 11 Q35 comparison table**

WLAN requires fixed infrastructure (APs) — MANET needs NONE. In WLAN, devices are end-users only — in MANET, every device is BOTH user AND router (multi-hop). WLAN has single point of failure (AP) — MANET is fully distributed. WLAN best for permanent locations; MANET for emergency/infrastructure-less environments.

Q5b [Exam]: Q5(b) — 3 marks: THREE applications of MANETs.

✅ **Answer — See Module 11 Q40 full answers**

(1) Tactical Military Networks — communication without fixed infrastructure in the field. (2) Emergency/Disaster Recovery — instant network after disasters (e.g., Haiti 2010). (3) Vehicle-to-Vehicle (V2V) communication — real-time road safety sharing.

Q5c [Exam]: Q5(c) — 8 marks: Explain FOUR constraints of MANETs.

✅ **Answer — See Module 11 Q41 full answer**

(1) Infrastructure-less — all networking functions distributed across devices; no central authority. (2) Dynamic topologies — constant movement causes route breakage, network partitions. (3) Physical layer limits — limited range, high packet loss, broadcast interference. (4) Limited resources — battery drain from routing, limited CPU and memory.

Q5d [Exam]: Q5(d) — 5 marks: Which routing protocol for effective MANET bandwidth utilization? Justify.

✅ **Answer — See Module 11 Q42 full justification**

Best: AODV (Reactive/On-demand). Justification: generates routing traffic ONLY when data needs to be sent → zero overhead during silence → entire bandwidth preserved for actual data. Proactive protocols (DSDV) continuously flood updates even when no data flows — wastes scarce wireless bandwidth.

EXAM SECTION

Test Paper 2 — Full Solved

Mobile Computing | DATE: 20th February 2026 | TIME: 30 minutes

Section 1: Multiple Choice — Answer Key with Explanations

Q	Question Summary	Answer	Why Correct
i	What does a deployment model primarily describe?	B — Location and management of cloud infrastructure	Describes WHERE and WHO manages — not what is built, how programmed, or billing method.
ii	Organization uses on-premises + bursts to public cloud during peak — this is?	C — Hybrid Cloud	Uses BOTH private (on-premises) AND public (burst) together = Hybrid. 'Bursting' is classic Hybrid Cloud pattern.
iii	Primary disadvantage of Private vs Public Cloud?	C — Higher upfront capital expenditure	Private Cloud requires buying/installing own hardware — large capital cost. Public = zero upfront, pay-as-you-go.
iv	Which 5G feature is most critical for MEC ultra-low latency?	C — Control and User Plane Separation (CUPS)	CUPS allows User Plane at the edge near user → minimal data travel distance → 1ms latency. Makes MEC possible.
v	Where does AI inferencing occur in MEC?	B — Edge Node/Server	Inferencing runs at nearby edge node → sub-millisecond response. Not distant cloud (50ms+).
vi	Primary driver for moving compute to edge in MEC?	B — Reduce network congestion and round-trip latency	Core reason: cut latency and congestion. Not storage, not eliminating base stations.
vii	Term used interchangeably with Serverless computing?	C — Function as a Service (FaaS)	Serverless = FaaS. Code runs as individual functions triggered by events. AWS Lambda is the example.

viii	Which IoT layer senses and gathers data?	C — Perception Layer	Perception (Sensing) Layer = bottom layer with all sensors, actuators, cameras, RFID devices.
ix	Significant security risk for IoT?	B — Hardcoded default passwords and lack of encryption	Default passwords + no encryption = millions of exploitable devices. Caused Mirai Botnet (2016).
x	A Chatbot is an application of which AI?	B — Natural Language Processing (NLP)	Chatbots understand and generate HUMAN LANGUAGE — the core task of NLP.

Section 2: True/False — Complete Answers with Corrections

#	Statement	T/F	Explanation & Correction
i	Reactive routing protocols have higher control overhead than proactive protocols.	F	FALSE. PROACTIVE protocols have higher overhead (constant table updates). REACTIVE = lower overhead (routes only on demand).
ii	AODV is an example of a proactive routing protocol.	F	FALSE. AODV = REACTIVE ('On-demand' is in the name). Proactive examples: DSDV, OLSR.
iii	ZRP uses proactive approach for nearby nodes and reactive for distant nodes.	T	TRUE. ZRP is HYBRID — proactive within local zone, reactive outside.
iv	Unicast in MANET = message from one source to all other nodes.	F	FALSE. Unicast = one source to ONE destination. One to ALL = Broadcast. One to a group = Multicast.
v	Network Slicing allows one physical 5G network → multiple virtual networks.	T	TRUE. Network Slicing creates isolated virtual networks on shared physical 5G infrastructure.
vi	Massive MIMO uses a	F	FALSE. Massive MIMO

	very small number of antennas to save space.		uses 64–256+ antennas. 'Massive' = massively LARGE number.
vii	Beamforming broadcasts signal in all directions equally for maximum coverage.	F	FALSE. Beamforming FOCUSES signal like a SPOTLIGHT at a specific user — opposite of omnidirectional.
viii	UPF in 5G handles signaling and authentication.	F	FALSE. UPF = USER DATA TRAFFIC. Signaling/auth = AMF (and AUSF). 'User Plane' ≠ signaling.
ix	MME is the primary 4G component handling actual user data traffic.	F	FALSE. MME = SIGNALING (auth, session control). User data = SGW (routing) + PGW (gateway to internet).
x	CUPS allows control logic to scale independently from data traffic.	T	TRUE. CUPS separates control plane from user plane — each scales independently. Enables edge deployment.

📝 **Exam Tip:** TRUE/FALSE EXAM STRATEGY: Watch for absolute words: 'all', 'only', 'very small', 'very large', 'primary', 'always', 'never'. These are often the clue that a statement is FALSE. Always write a one-sentence correction for any FALSE answer to earn full marks.

EXAM STRATEGY — 4-Step Winning Formula & Quick Reference Tables

Question Word	What It Means	Steps Required	Mark Expectation
Define	Give the meaning.	Step 1 only — be direct.	~40% if only definition given
Explain	Describe how it works.	Steps 1+2+3 (example essential).	~80%+ with define+explain+example
Discuss	Talk about in depth.	All 4 steps.	Full marks possible
Compare	Show differences and similarities.	Steps 1+2 for EACH — always use a TABLE.	Use a comparison table always
Justify	Give reasons why something is chosen.	Steps 2+3+4 — show evidence.	Recommendation + clear justification
True/False	Identify if statement is correct.	State T or F + one-sentence correction if false.	Full mark requires correction sentence

📝 **Exam Tip:** KEY RULE: 'Explain', 'Discuss', 'Justify' — ALWAYS include a real-world example. Students who only define get ~40%. Students who define + explain + real example get 80%+.

Quick Reference — Key Numbers & Formulas

Fact	Value / Detail
Throughput formula	$\eta = P / (T \times C)$ — packet size / (time × channel capacity)
Wireless BER	~10^{-3} (vs wired 10^{-6} — wireless is 1,000× worse)
GSM channel structure	200 kHz FDMA × 8 TDMA slots = 8 users per frequency channel
GSM SMS max length	160 alphanumeric characters
S-MAC active period	Fixed at 115ms
GPRS max speed	171.2 kbps theoretical; 5–40 kbps realistic
UMTS (3G) base speed	384 kbps downlink; HSDPA: 3.6+ Mbps
4G LTE typical speed	~50–150 Mbps
5G peak speed	Up to 20 Gbps (100× faster than 4G)
5G latency	As low as 1ms (50× lower than 4G's 30–50ms)

5G device density	1 million devices/km² (250× more than 4G's ~4,000)
GPRS QoS Delay Class 1	<0.5 seconds end-to-end
WSN protocol overhead	Control traffic can be up to 70% of total traffic
Mirai Botnet scale	600,000 infected IoT devices (2016 — default password attack)

Generation Comparison — 1G to 5G

Gen	Standard	Technology	Key Services	Max Speed
1G	AMPS, NMT	Analog FDMA	Voice only	N/A — analog
2G	GSM	Digital TDMA+FDMA	Voice, SMS, basic data	9.6 kbps
2.5G	GPRS	Packet-switched over GSM	Mobile internet, email	~40–171 kbps
3G	UMTS	W-CDMA	Video calls, broadband data	384 kbps–3.6 Mbps
4G	LTE	OFDMA, all-IP EPS	HD video, fast internet	50–150 Mbps
5G	NR (New Radio)	mmWave + Sub-6GHz, OFDM	AR/VR, autonomous vehicles, IoT	Up to 20 Gbps

Good luck in your examination!

Define → Explain → Real Example → Conclude. Always use tables for comparisons.